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Self-capacitive sensor apparatus

Abstract

An apparatus for use at an inventory location stowing items comprises an array of sensor elements. Each sensor element comprises a plurality of conductive elements that are driven as a self-capacitance sensor. Based on changes in capacitance values, interactions between a lane and user may be determined. The conductive elements may be selectively addressed to provide various physical resolutions and operational configurations. Some geometries of conductive elements provide information about position of an interaction along a long axis of the lane, as well as presence of items in the lane. The sensor element may include three conductors, stacked atop one another and separated by an insulator. To compensate for temperature and other effects, these the outermost conductors are grounded, and innermost conductor is driven. The array may be implemented on a rigid or flexible substrate. The array operates reliably in the presence of water condensation or small spills.

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Background/Summary

BACKGROUND

(1) Product distributors typically maintain an inventory of various items that may be ordered, purchased, leased, borrowed, rented, viewed, and so forth, by clients or customers. For example, an e-commerce website may maintain inventory in a fulfillment center. When a customer orders an item, the item is picked from inventory, routed to a packing station, packed, and shipped to the customer. Likewise, physical stores maintain inventory in customer accessible areas, such as in a shopping area, and customers can pick items from inventory and take them for purchase, rental, and so forth.
(2) Other examples of entities that maintain facilities holding inventory include libraries, museums, rental centers, and so forth. In each instance, for an item to be moved from one location to another, it is picked from its current location and transitioned to a new location. It is often desirable to monitor quantity or movement of inventory within the facility.

Description

BRIEF DESCRIPTION OF FIGURES

(1) The detailed description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items or features. The figures are not necessarily drawn to scale, and in some figures, the proportions or other aspects may be exaggerated to facilitate comprehension of particular aspects.
(2) FIG. **1** illustrates a system using an apparatus comprising capacitive sensors to generate interaction data about an inventory location, according to some implementations.
(3) FIG. **2** illustrates a block diagram of the apparatus, according to some implementations.
(4) FIG. **3**A illustrates a first arrangement of conductive elements of the sensor elements of the apparatus, according to some implementations.
(5) FIG. **3**B illustrates a second arrangement of conductive elements of the sensor elements of the apparatus including a compensation sensor element, according to some implementations.
(6) FIG. **4** illustrates a third arrangement of conductive elements of the sensor elements of the apparatus, according to some implementations.
(7) FIG. **5** illustrates a fourth arrangement of conductive elements of the sensor elements of the apparatus, according to some implementations.
(8) FIG. **6** illustrates a fifth arrangement of conductive elements of the sensor elements of the apparatus, according to some implementations.
(9) FIG. **7** illustrates operation of the apparatus with an item management accessory, according to some implementations.
(10) FIG. **8** illustrates the apparatus with an extended cover layer, according to some implementations.
(11) FIG. **9** illustrates several layouts of sensor elements of the apparatus, according to some implementations.
(12) FIG. **10** illustrates operation of triangular sensor elements of the apparatus, according to some implementations.
(13) FIG. **11** is a graph of capacitance over time depicting various interactions, according to some implementations.
(14) FIG. **12** depicts a block diagram of a first implementation to determine interaction data.
(15) FIG. **13** depicts a block diagram of a second implementation to determine interaction data.

(16) FIG. **14** depicts graphs of capacitance and an interrupt-based technique to accumulate item count over time responsive to various interactions, according to some implementations.
(17) FIG. **15** depicts graphs of capacitance and a polling-based technique to accumulate item count over time responsive to various interactions, according to some implementations.
(18) FIG. **16** is a flow diagram of a process for determining the association between sensor elements and lanes at an inventory location, according to some implementations.
(19) FIG. **17** depicts active and inactive sensor elements associated with respective lanes, according to some implementations.
(20) FIG. **18**A depicts block diagrams of the conductive elements and respective connections for a first mode and a second mode, according to some implementations.
(21) FIG. **18**B depicts block diagrams of the conductive elements and respective connections for a first variation and a second variation of the second mode, according to some implementations.
(22) FIG. **19**A depicts conductive element configurations for a first conductive element arrangement in the first mode and the second mode, according to some implementations.
(23) FIG. **19**B depicts conductive element configurations for a second conductive element arrangement in the first mode and a first variation of the second mode, according to some implementations.
(24) FIG. **19**C depicts conductive element configurations for the second conductive element arrangement in a second variation of the second mode, according to some implementations.
(25) FIG. **20** depicts conductive element configurations for a second conductive element arrangement in the first mode and the second mode, according to some implementations.
(26) FIG. **21** is a flow diagram of a process for using the apparatus to determine compensation data, according to some implementations.
(27) FIG. **22** is a flow diagram of a process for combining operation of the sensor elements to determine proximity of an object, according to some implementations.
(28) FIG. **23** is a flow diagram of a process for using a first sensor element proximate to a second sensor element to determine a type of interaction, according to some implementations.
(29) While implementations are described herein by way of example, those skilled in the art will recognize that the implementations are not limited to the examples or figures described. It should be understood that the figures and detailed description thereto are not intended to limit implementations to the particular form disclosed but, on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean “including, but not limited to”.

DETAILED DESCRIPTION

(30) Described in this disclosure are systems and techniques for generating interaction data at an inventory location that may be used in a materials handling facility (facility). The facility may include, or have access to, an inventory management system. The inventory management system may be configured to maintain information about items, users, condition of the facility, and so forth. For example, the inventory management system may maintain data indicative of a number of items at a particular inventory location, what items a particular user is ordered to pick, how many items have been picked or placed at the inventory location, requests for assistance, environmental status of the facility, and so forth.
(31) Operation of the facility may be facilitated by using one or more sensors to acquire information about interactions in the facility. The inventory management system may process the sensor data from the one or more sensors to determine interaction data. The interaction data is indicative of an action such as picking or placing an item at a particular inventory location, presence of the user at the inventory location, and so forth. For example, the inventory management system may use the sensor data to generate interaction data that determines a type of item a user picked from a particular inventory location.
(32) An inventory location may include shelves, hangers, and so forth, or a portion thereof that holds or otherwise supports a type of item. The inventory location may comprise a section, such as individual lanes, of a larger inventory location, such as a shelf. For example, a shelf may have eight lanes, with each lane holding a specified type of item. Items may be added to (placed) or removed (picked) from the inventory location, moved from one inventory location to another, and so forth.
(33) An apparatus comprising a plurality of capacitive sensors may be used at or in conjunction with an inventory location. For example, the apparatus may comprise a mat that rests on a shelf or may be integrated into the shelf. In some implementations the inventory location may be in view of sensor range of other sensors such as image sensors. An inventory management system may use the data from these sensors to determine the interaction data.
(34) The capacitive sensors of the apparatus comprise electronics and a plurality of sensor elements, each sensor element in turn comprising a plurality of conductive elements. Objects on or above the sensor elements may result in a particular electrical capacitance of the particular sensor element. For example, the presence of an item may result in a first capacitance value while the absence of the item may result in a second capacitance value. Given a previously known arrangement of the sensor elements with respect to the apparatus, an estimated location of an interaction or an object may be determined. For example, a particular set of sensor elements may be associated with lane **1**, and a change in the capacitance as measured by that particular set of sensor elements may indicate an interaction has taken place at that lane. The interaction may include an item being picked from or placed to the inventory location, presence of a hand of the user, presence of a manipulator of a robot, and so forth.
(35) The sensor elements, and their respective conductive elements, may comprise a variety of different arrangements. In one arrangement, individual sensor elements may be approximately rectangular and arranged such that their long axis extends from a front of the apparatus to a back of the apparatus. One or more individual sensor elements may be associated with a particular lane. Based on a change in capacitance, an interaction may be associated with a particular lane. In another arrangement, the sensor elements may be arranged as right triangles, with two triangles arranged such that their respective longest sides are adjacent to one another. A location of an object may be determined with respect to these triangular conductive elements based at least in part on a ratio or difference between the capacitance values obtained by each. For example, the first capacitance value obtained using a first triangular conductive element and a second capacitance value obtained using a second triangular conductive element may be approximately the same when a homogenous object is located proximate to the midpoints of the longest sides of each triangle.
(36) The sensor elements may be configured to utilize self-capacitance of the conductive elements, rather than a mutual capacitance. In one implementation, a known charge may be provided to the conductive element, and the resultant voltage may be measured between the conductive element and the ground. An active shield comprising an electrical conductor may be arranged along one or more sides of the conductive element. For example, the shield may be separated from the conductive element by an electrical insulator. During operation, the shield may be driven to the same, (or a substantially similar) electrical potential as that provided to the conductive element. As a result of this, a voltage difference that is below a threshold voltage is present between the shield and the conductive element. In some implementations, the voltage difference may be zero. The shield in this configuration may direct the electric field generally away from the shield. This directionality may be used to prevent erroneous readings for objects on the back side of the conductive element, such as may occur in an unshielded configuration. In some implementations, a ground plane may be arranged behind the shield, opposite the conductive element. The ground plane may comprise an electrically conductive material that is separated from the shield by an electrical insulator. The ground plane may be connected to an earth ground in some implementations. Proximity of an object to

the conductive element or contact by the object to the conductive element affects the charge on the conductive element, producing a change in the resultant voltage that may then be measured and used to determine a capacitance value.
(37) Each sensor element may comprise a plurality of conductive elements in various arrangements. These arrangements allow for various options with regard to fabrication and deployment. In a first arrangement, three conductive elements are stacked vertically, and separated by an electrical insulator. In a second arrangement, the apparatus comprises two assemblies that are joined for operation: a first assembly that comprises two conductive elements stacked vertically, and a second assembly that adds a third conductive element. In the second arrangement, some sensor elements may comprise only two conductive elements, as provided by the first assembly, while some sensor elements may comprise the two conductive elements plus the third afforded by the added second assembly. In other arrangements, various combinations of different widths and relative spacing between the conductive elements may be used. Other arrangements of the conductive elements are also described.
(38) The sensor elements, and their respective electrical conductors, are connected to control circuitry using a plurality of conductors. The control circuitry, plurality of conductors, and sensor elements may be arranged such that the apparatus may be cut to allow customized widths (left to right) and depths (front to back) to be achieved. For example, the apparatus may be trimmed from 1.2 meters wide to 1 meter wide to fit a shelf.
(39) During operation, a switch module may comprise circuitry used to switch or selectively connect a capacitive sensor module to a particular conductive element. Once connected, a charge may be applied to the conductive element. A voltage, capacitance, resistance, or other electrical characteristic may be measured by the capacitive sensor module which may then generate capacitance data.
(40) The switch circuitry may be used to operate various configurations of conductive elements and provide different operational modes. In a first mode, or sensing mode, the sensor element is operated to determine capacitance of conductive elements that provide an electrical field that would impinge on a load of the apparatus, such as within a lane where an item may be stowed. In a second mode, or compensation mode, the sensor element is operated to determine capacitance between two conductive elements of the sensor element, allowing for compensation due to variations resulting from manufacture, ambient temperature, temperature of an item stowed at the apparatus, and so forth.
(41) The switch circuitry may be operated to provide various scan modes. In an individual scan mode, individual sensor elements and their respective conductive elements are connected and used to determine the capacitance value of the sensor element. In a selective scan mode, previously determined sets of sensor elements and their respective conductive elements are connected and used to determine the capacitance value of the respective sets. In a global mode the switch circuitry may connect to all of the sensor elements to determine a capacitance value that is then used to determine proximity of an object, such as a user approaching the apparatus.
(42) An event detection module may be used to generate event data associated with a particular inventory location. Event data may comprise information that is indicative of a change associated with an inventory location. For example, the event data may indicate that one or more types of sensor data have exceeded particular thresholds or met other conditions.
(43) The event data may be used to trigger or initiate other actions, such as determination of interaction data. For example, event data may indicate occurrence of a change in capacitance values that exceeds a threshold value, change relative to a previously set baseline value, and so forth. Responsive to this, interaction data indicative of a pick or place of an item at the inventory location may be determined. In another example, the event data may be used to determine a time that an interaction took place. This time may then be used to determine other sensor data, such as from an image sensor or a weight sensor, that may be used to determine the interaction data.
(44) By using the techniques described herein, operation of the facility may be improved. Details about interactions between users and items in the facility may be quickly and accurately determined. For example, as items are picked, placed, and so forth, information such as inventory levels based on changes in the count of items at the inventory locations may be readily and more accurately determined. As a result, the inventory management system may be able to quickly track what item a user has interacted with, maintain up-to-date inventory information, and so forth.
(45) Illustrative System
(46) FIG. **1** illustrates a system **100** comprising an apparatus **102** at an inventory location that provides data that is used to determine interaction data, according to some implementations. The apparatus **102** be associated with one or more lanes **104**. The lanes **104** are specified areas upon or near the apparatus **102** that are associated with a particular type of item **106**. For example, the apparatus **102** depicted here has eight lanes **104**(**1**), **104**(**2**), . . . and **104**(**8**). A first lane **104**(**1**) may be associated with storing some quantity of a first item **106**(**1**) such as grape drink, while a second lane **104**(**2**) may be associated with storing some quantity of a second item **106**(**2**) such as orange drink, and so forth. The apparatus **102** may be arranged adjacent to one another. For example, a separate apparatus **102** may be associated with individual shelves on a rack.
(47) A user **108** may interact with the lanes **104** associated with the apparatus **102**. For example, the user **108** may remove (pick) a first item **106**(**1**) from the first lane **104**(**1**) and place the first item **106**(**1**) into a tote **110**. The tote **110** may be associated with the user **108** or that user's **108** user account. Items **106** may also be added to (placed) the lanes **104** of the apparatus **102**. For example, a person or robot may restock the lane **104** with items **106**.
(48) Each apparatus **102** comprises one or more sensor elements **150**, and in some implementations may include or be associated with one or more other sensors **112**. The other sensors **112** may be positioned to gather information about one or more inventory locations, such as particular lanes **104** or the entire shelf. For example, the other sensors **112** may comprise one or more accelerometers, hygrometers, and so forth.
(49) Circuitry associated with the other sensors **112** generates other data **114**. For example, the other sensors **112** may comprise weight sensors arranged proximate to each of the four corners of the apparatus **102**. Output from each weight sensor **112** may be used to produce the other data **114** that includes weight values for each of the corners at particular times. In another example, the other sensors **112** may comprise an image sensor, such as a camera having a field of view (FOV) **120** that includes at least a portion of the apparatus **102**. Output from the image sensor **112** may be processed to determine the interaction data.
(50) The apparatus **102** comprises one or more sensor elements **150**(**1**)-(N). For example, the conductive elements of the sensor elements **150** may be located on, in, or below a surface of the apparatus **102** upon which items **106** may rest. Various structures and arrangements of the sensor elements **150** and their respective conductive elements are described in more detail in the following figures.
(51) The sensor elements **150** are coupled to an apparatus control module **132** comprising capacitive sensing circuitry that generates capacitance data **116**. The capacitive sensing circuitry may use various techniques to determine capacitance. For example, the capacitive sensing circuitry may include a source that provides a predetermined voltage, a timer, and circuitry to measure voltage of the conductive element relative to the ground. By determining an amount of time that it takes to charge the conductive element to a particular voltage, the capacitance may be calculated. The capacitive sensing circuitry may use one or more of analog or digital circuits to determine capacitance. During operation, one or more of the sensor elements **150** may be operated to determine capacitance data **116** indicating capacitance values at particular times and with respect to the different portions of the apparatus **102** associated with the set of sensor elements **150** that are operated.
(52) In some implementations, each lane **104** or group of lanes **104** may be associated with one or more sensor elements **150**. For example, each lane **104** may be associated with a single sensor element **150**. In another example, the lanes **104** may span two or more sensor elements **150**.
(53) An inventory management system **122** may access the capacitance data **116** and, in some implementations, the other data **114**. The inventory management system **122** may be configured, as described below, to perform various functions such as tracking changes to a quantity on hand of the items **106** at the apparatus **102**.
(54) The inventory management system **122** may include or have access to an analysis module **124**. The analysis module **124** may access information including, but not limited to, item data **126**, lane configuration data **130**, capacitance data **116**, event data, other data **114**, or other information.
(55) The item data **126** provides information about a particular type of item **106**, including characteristics of that type of item **106** such as physical dimensions, where that type of item **106** is located in the facility, characteristics about how the type of item **106** appears, capacitance values

associated with the type of item **106**, and so forth. For example, the item data **126** may indicate that the type of item **106** is "Brand X grape drink, 16 oz can" with a stock keeping unit (SKU) number of "24076513. The item data **126** may indicate the types and quantities of items **106** that are expected to be stored at that particular inventory location such as in a particular lane **104** on the apparatus **102**, width and depth of that type of item **106**, weight of the type of item **106** individually or in aggregate, sample images of the type of item **106**, and so forth.
(56) The item data **126** may include an item identifier. The item identifier may be used to distinguish one type of item **106** from another. For example, the item identifier may include a SKU string, Universal Product Code (UPC) number, radio frequency identification (RFID) tag data, and so forth. The items **106** that are of the same type may be referred to by the same item identifier. In other implementations, non-fungible items **106** may each be provided with a unique item identifier, allowing each to be distinguished from one another.
(57) The item data **126** may include one or more of geometry data, item weight data, sample image data, sample capacitance data, or other data. The geometry data may include information indicative of size and shape of the type of item **106** in one, two, or three dimensions. For example, the geometry data may include the overall shape of an item **106**, such as a cuboid, sphere, cylinder, and so forth. The geometry data may also include information such as length, width, depth, and so forth, of the type of item **106**. Dimensional information in the geometry data may be measured in pixels, centimeters, inches, arbitrary units, and so forth. The geometry data may be for a single item **106**, or a package, kit, or other grouping considered to be a single item **106**.
(58) The item weight data comprises information indicative of a weight of a single item **106**, or a package, kit, or other grouping considered to be a single item **106**. The item data **126** may include other data. For example, the other data may comprise weight distribution of the item **106**, point cloud data for the item **106**, and so forth.
(59) The sample capacitance data may comprise data indicative of a previously measured or calculated change in capacitance of a representative sensor element **150** based on the presence or absence of a sample of the type of item **106**. For example, during processing or intake of the item **106** at the facility, a sample of the type of item **106** may be placed on a sensor element **150** to generate the sample capacitance data.
(60) The sample image data may comprise one or more images of one or more of that type of item **106**. For example, sample image data may be obtained during processing or intake of the item **106** to be used by the facility.
(61) The item data **126** may include one or more inventory location identifiers (IDs). The inventory location ID is indicative of a particular area or volume of an inventory location such as an apparatus **102** that is designated for stowage of the type of item **106**. For example, a single apparatus **102** may be associated with several lanes **104**, each with a different inventory location ID. Each of the different inventory location IDs may be associated with a lane **104** having a particular area on the apparatus **102** designated for storage of a particular type of item **106**. A single type of item **106** may be associated with a particular inventory location ID, a plurality of inventory location IDs may be associated with the single type of item **106**, more than one type of item **106** may be associated with the particular inventory location ID, and so forth.
(62) The item data **126** may also include quantity data. The quantity data may comprise a count or value indicative of a number of items **106**. The count may be a measured or an estimated value. The quantity data may be associated with a particular inventory location ID, for an entire facility, and so forth. For example, the same type of item **106** may be stored at different shelves **104** within the facility. The quantity data may indicate the quantity on hand for each of the different inventory locations.
(63) The lane configuration data **130** may comprise information that associates particular sensor elements **150** of a particular apparatus **102** with a particular lane **104**. For example, the first lane **104**(**1**) of the apparatus **102** depicted may be associated with sensor element **150**(**2**), while the eighth lane **104**(**8**) of the same apparatus **102** depicted is associated with sensor elements **150**(**17**)-**150**(**19**).
(64) The analysis module **124** may utilize one or more of the capacitance data **116**, event data, the other data **114**, the item data **126**, the lane configuration data **130**, or other information to generate interaction data **128**. The interaction data **128** is indicative of an action such as picking or placing an item **106** for a particular inventory location, presence of the user **108** at the inventory location, and so forth. Operation of the analysis module **124** is described in more detail below with regard to FIG. **12** and elsewhere.
(65) FIG. **2** illustrates at **200** a block diagram of the apparatus **102**, according to some implementations. The apparatus control module **132** may comprise a control module **202**, a capacitive sensor module **206**, a switch module **212**, and a plurality of sensor elements **150**(**1**)-(N). The apparatus control module **132** comprises electronic circuitry coupled together using electrical conductors, such as wires, circuit traces, and so forth. The electronic circuitry may comprise one or more processors, controllers, microcontrollers, programmable logic devices, and so forth that execute one or more instructions.
(66) The control module **202** may generate sensor control data **204** that operates the capacitive sensor module **206**. The capacitive sensor module **206** may comprise a capacitive signal source that generates a capacitive signal **208**. During operation, the capacitive signal **208** may be used to supply a charge to one or more conductive elements **250** as distributed via the switch module **212**. The capacitive sensor module **206** may comprise a shield signal source that generates a shield signal **210**. During operation, the shield signal source may generate the shield signal **210** that is at the same voltage potential of the input of the capacitive signal **208**. The shield signal **210** may then be used to supply a charge to the one or more conductive elements **250** as distributed with the switch module **212**. This provides an actively driven shield, or active shield, in which there is no difference in electrical potential between the conductive elements **250** supplied with the capacitive signal **208** and the conductive elements **250** supplied with the shield signal **210**. External interference may then couple to the conductive elements **250** driven as the active shield, producing little interaction with the conductive elements **250** being supplied by the capacitive signal **208**. Using this technique, the sensor element **150** may detect objects on the side opposite that of the conductive element **250** driven by the shield signal **210**, with that conductive element **250** preventing the sensor from "seeing" or being affected by an object behind the shield. The active driving technique also substantially improves the signal to noise ratio of the resulting capacitance data **116**, improving overall performance.
(67) In some implementations, the actively driven shield may be omitted. For example, the actively driven shield may be omitted if the overall thickness T**34** of the apparatus **102**, such as shown in FIG. **3**A, exceeds a threshold amount.
(68) The capacitive sensor module **206** may be operated in one or more different sensitivity modes. Different sensitivity modes may be characterized by one or more of different amplitudes of the capacitive signal **208**, different integration times, different frequencies of operation, and so forth. For example, a high sensitivity mode may utilize a first amplitude of capacitive signal **208** that is greater than a second amplitude that is associated with a low sensitivity mode. While operating in the high sensitivity mode, the range of the electric field associated with operation may be extended. For example, the high sensitivity mode may be used to determine the approach of a user **108** or other object to the apparatus **102**, while the low sensitivity mode is used to detect and characterize the pick or place of items **106** to an inventory location. The same sensor element **150** or set of sensor elements **150** may be operated at different sensitivity modes at different times. For example, at a first time a first set of sensor elements **150** may be operated at high sensitivity, and at a second time the same first set of sensor elements **150** may be operated at low sensitivity. The operating mode used may be dynamically adjusted. For example, a high sensitivity mode may be used to determine an approach of a user **108**, and once detected, the apparatus **102** may transition to using the low sensitivity mode.
(69) The control module **202** may also generate switch control data **214** that operates the switch module **212** to connect particular conductive elements **250** to particular portions of the capacitive sensor module **206**. The switch module **212** comprises switch circuitry, such as one or more relays, transistors, or other devices that allow for the selective connection of conductive elements **250** to particular portions of the capacitive sensor module **206**, the ground **248**, and so forth. For example, the switch module **212** may selectively connect particular conductive elements **250** to the capacitive signal source, the shield signal source, ground **248**, and so forth. Various switch configurations are discussed in the following figures. In some implementations, the ground **248** may provide a relatively high impedance path to a chassis ground, earth ground, and so forth.
(70) The switch module **212** is connected to the respective conductive elements **250** of the sensor elements **150**(**1**)-(N) by a plurality of conductors

270. In one implementation the plurality of conductors **270** may comprise electrically conductive traces on a rigid or flexible substrate. In some implementations, the apparatus **102** may utilize one or more multiplexing techniques, additional switch modules **212**, and so forth.

(71) In implementations in which compensation sensor elements **372** (see FIG. **3**B) are present, the switch module **212** is connected to the respective conductive elements **250** of the compensation sensor elements **372**(**1**)-(D) by a plurality of conductors **270**. In one implementation, the plurality of conductors **270** may comprise electrically conductive traces on a rigid or flexible substrate. In some implementations, the apparatus **102** may utilize one or more multiplexing techniques, additional switch modules **212**, and so forth.

(72) The capacitive sensor module **206** provides as output raw capacitance data **252**. The raw capacitance data **252** may comprise data, such as values indicative of capacitance with respect to the conductive elements **250** that are coupled to the capacitive signal source by the switch module **212** at a particular time. While operating in a first mode **1802**, such as described below with regard to FIG. **18**, the raw capacitance data **252** may comprise load capacitance data **254**, indicative of the capacitance with respect to a portion of the sensor elements **150** that is associated with where items **106** may be stored. In comparison, while operating in a second mode **1804**, such as described below with regard to FIG. **18**, the raw capacitance data **252** may comprise reference capacitance data **256**, indicative of the capacitance between particular conductive elements **250**.

(73) In some implementations, compensation sensor elements **372** (see FIG. **3**B) may be used to determine the reference capacitance data **256**. For example, the compensation sensor elements **372** may be operated to determine reference capacitance data **256** while the sensor elements **150** are operated to determine the load capacitance data **254**.

(74) A compensation module **258** may accept as input the raw capacitance data **252** and determine as output the capacitance data **116**. In one implementation the compensation module **258** may subtract the reference capacitance data **256** from the load capacitance data **254** to determine the capacitance data **116**. In another implementation the compensation module **258** may, based on the reference capacitance data **256**, determine a correction factor. The load capacitance data **254** may then be multiplied by the correction factor to determine the capacitance data **116**. In other implementations, other techniques may be used to determine the capacitance data **116**.

(75) In some implementations, operation of the compensation module **258** may be initiated based on other data **114**. For example, based on an other sensor **112** comprising a hygrometer indicating a change in humidity that exceeds a threshold value, the compensation module **258** may be operated to update the reference capacitance data **256**. In another example, based on an other sensor **112** such as an accelerometer indicating the apparatus **102** has been moved, the compensation module **258** may be operated to update the reference capacitance data **256**.

(76) In some implementations, the apparatus control module **132** may operate in a serial fashion, operating different combinations of sensor elements **150**, compensation sensor elements **372**, and so forth in series. For example, during a specified time interval, a particular set of sensor elements **150** or compensation sensor elements **372** may be operated to determine raw capacitance data **252**.

(77) The apparatus control module **132** may include a communication interface **260**. The communication interface **260** may comprise one or more wired or wireless communication interfaces. For example, the communication interface **260** may comprise a WiFi communication interface to provide wireless communication with another device, such as the inventory management system **122**.

(78) In some implementations external control data **280** from an external source may be received by the communication interface **260** of the apparatus control module **132**. For example, the inventory management system **122** may send external control data **280** specifying one or more parameters for operation of the apparatus control module **132**, requesting capacitance data **116**, and so forth.

(79) FIG. **3**A illustrates at **300** a first arrangement of conductive elements **250** of the sensor elements **150** of the apparatus **102**, according to some implementations. A top view **302** of an apparatus **102** and a side view **304** of an enlarged portion of the apparatus **102** are depicted. The views are shown with respect to three mutually orthogonal axes. Relative to the apparatus **102** while placed in a generally flat orientation for use, the X axis extends left-to right (width), the Y axis extends back to front (depth) and the Z axis extends bottom to top (thickness).

(80) As shown in the top view **302**, a plurality of sensor elements **150**(**1**)-(N) are distributed in a single row of columns across the apparatus **102**, forming an array. The apparatus control module **132** is located in a corner, and the conductors **270** connect the conductive elements **250** of the sensor elements **150** to the apparatus control module **132**.

(81) The sensor elements **150** may comprise various configurations of conductive elements **250**. In one implementation, an expanded conductive element **310** may comprise a plurality of conductive elements that are electrically connected to operate as a single conductive element. In comparison, in another implementation a unitary conductive element **312** may comprise a single contiguous conductive element.

(82) In this illustration, the expanded conductive element **310** is depicted as having four electrically conductive traces that are electrically connected. In other implementations, other configurations may be used. Other examples of the expanded conductive element **310** such as a mesh pattern, hash pattern, hatch pattern, comb pattern, fractal pattern, and so forth may be used. In some implementations, an expanded conductive element **310** may comprise one or more electrical conductors that are electrically connected before connection to the switch module **212**. This electrical connection may be distinct from the connectivity provided by the switch module **212**. For example, the electrical connection between portions of the expanded conductive element **310** may be permanent.

(83) The expanded conductive element **310** may have a conductor coverage ratio that is less than one. In one implementation the conductor coverage ratio may be calculated as a sum of the area of electrical conductors divided by the total area described by a boundary that encompasses the portions of the expanded conductive element **310**. For example, a boundary that extends around the perimeter of a conductive element **250** may result in an area of 10 square millimeters. Within that 10 square mm, the expanded conductive element **310** may consist of four traces, each trace having a respective surface area of 1 square mm. As a result, the conductor coverage ratio in this example would be (1+1+1+1)/10=0.40.

(84) Unless otherwise specified, the individual conductive elements **250** as used in this disclosure, such as with regard to sensor elements **150** or compensation sensor elements **372** (described with regard to FIG. **3**B), may comprise expanded conductive elements **310**, unitary conductive elements **312**, or combinations thereof.

(85) The arrangement of the apparatus control module **132**, the sensor elements **150** and their respective conductive elements **250**, and the conductors **270** may facilitate cutting the apparatus **102** along one or more of a width cut line **322** or a depth cut line **324**. For example, the apparatus **102** may be cut to fit within a specified shelf.

(86) The side view **304** depicts an enlarged portion of a cross section of the apparatus **102** along line "A**1**-A**1**". In the implementation shown, a cover layer **306** is provided. The cover layer **306** may comprise a dielectric material, such as non-conductive plastic, ceramic, and so forth. The sensor elements **150** are arranged beneath the cover layer **306**. The cover layer **306** has a thickness T**31**. In some implementations, the cover layer **306** may comprise a portion of an electrical insulator **308**.

(87) In this implementation, each sensor element **150** comprises three conductive elements **250**(**1**)-(**3**) that are stacked vertically with respect to one another. Each conductive element **250** in this illustration consists of a respective expanded conductive elements **310** comprising four electrical conductors that are electrically connected. Each of the conductive elements **250**(**1**)-(**3**) have an overall width W**31** and thickness T**32** that are the same. In one implementation, the width W**31** may be 20 mm. The conductive elements **250**(**1**)-(**3**) are arranged with the same spacing T**33** between the conductive elements **250** of the same sensor element **150**.

(88) An overall thickness T**34** is also shown extending from an uppermost portion of the conductive element **250**(**1**) to a bottommost portion of the insulator **308**.

(89) In some implementations the conductive elements **250** may comprise one or more of metal, electrically conductive ink, electrically conductive plastic, and so forth. For example, a conductive element **250** may comprise a copper wire that is in contact with a region printed with conductive ink.

(90) An electrical insulator ("insulator") **308** is disposed between and underneath the conductive elements **250**(**1**)-(**3**). Respective ones of the conductors **270** (not shown) are coupled to respective ones of the conductive elements **250**(**1**)-(**3**). A spacing width of W**32** is maintained between

respective sensor elements **150**.
(91) In the implementation depicted, the apparatus **102** may be fabricated in various ways. For example, the conductive elements **250** and insulators **308** may be arranged in layers during fabrication and then joined together in a single assembly. In another example, individual layers, such as with conductive elements **250** deposited thereon, may be fabricated, and then stacked to provide the single assembly.
(92) FIG. **3**B illustrates at **350** a second arrangement of conductive elements **250** of the sensor elements **150** of the apparatus **102** including a compensation sensor element, according to some implementations. A top view **352** of an apparatus **102** and a side view **354** of an enlarged portion of the apparatus **102** are depicted. The views are shown with respect to three mutually orthogonal axes. Relative to the apparatus **102** while placed in a generally flat orientation for use, the X axis extends left-to right (width), the Y axis extends back to front (depth) and the Z axis extends bottom to top (thickness).
(93) As shown in the top view **352**, a plurality of sensor elements **150**(**1**)-(N) are distributed in a single row of columns across the apparatus **102**, forming an array. The apparatus control module **132** is located in a corner, and the conductors **270** connect the conductive elements **250** of the sensor elements **150** to the apparatus control module **132**.
(94) In some implementations, compensation sensor elements **372** may be used to determine the reference capacitance data **256** or other input to the compensation module **258**. A plurality of compensation sensor elements **372**(**1**)-(**3**) are shown in FIG. **3**B. In this illustration, the compensation sensor elements **372** are distributed to the left, center, and right, of the array of sensor elements **150**(**1**)-(N). In other implementations the compensation sensor element(s) **372** may be placed in other locations with respect to the apparatus **102**.
(95) As described above, the sensor elements **150** may comprise various configurations of conductive elements **250**. In the implementation depicted, each conductive element **250** of the sensor elements **150** comprises an expanded conductive element **310**. As mentioned above, the expanded conductive element **310** may have a conductor coverage ratio that is less than one. In other implementations, other configurations may be used.
(96) As described above, the arrangement of the apparatus control module **132**, the sensor elements **150** and their respective conductive elements **250**, the compensation sensor elements **372** and their respective conductive elements **250**, and the conductors **270** may facilitate cutting the apparatus **102** along one or more of a width cut line **322** or a depth cut line **324**. For example, the apparatus **102** may be cut to fit within a specified shelf.
(97) The side view **354** depicts an enlarged portion of a cross section of the apparatus **102** along line “A**2**-A**2**”. In the implementation shown, the cover layer **306** such as described above is shown. The cover layer **306** has a thickness T**35**.
(98) In this implementation, each sensor element **150** comprises four conductive elements **250**(**2**)(**1**)-(**2**)(**4**) that are stacked vertically with respect to one another. Each conductive element **250** in this illustration is a respective expanded conductive element **310**. Each of the conductive elements **250**(**2**)(**1**)-(**2**)(**4**) have an overall width W**36** and thickness T**36** that are the same. In one implementation, the width W**36** may be 20 mm. The conductive elements **250**(**2**)(**1**)-(**2**)(**4**) may be arranged with the same spacing T**37** between the conductive elements **250** of the same sensor element **150**.
(99) In this implementation, each compensation sensor element **372** comprises four conductive elements **250**(**1**)(**1**)-(**1**)(**4**) that are stacked vertically with respect to one another. Each conductive element **250** in this illustration consists of a respective expanded conductive element **310** comprising four electrical conductors that are electrically connected. Each of the conductive elements **250**(**1**)(**1**)-(**1**)(**4**) have an overall width W**35** and thickness T**38** that are the same. In one implementation, the width W**35** may be 6 mm. The conductive elements **250**(**1**)(**1**)-(**1**)(**4**) may be arranged with the same spacing T**37** between the conductive elements **250**. The width W**35** of the compensation sensor element **372** may be less than the width W**36** of the sensor element **150**.
(100) In some implementations the conductive elements **250** may comprise one or more of metal, electrically conductive ink, electrically conductive plastic, and so forth. For example, a conductive element **250** may comprise a copper wire that is in contact with a region printed with conductive ink.
(101) An electrical insulator (“insulator”) **308** is disposed between and underneath the conductive elements **250**(**1**)-(**3**). Respective ones of the conductors **270** (not shown) are coupled to respective ones of the conductive elements **250**(**1**)-(**3**). A spacing width of W**32** is maintained between respective sensor elements **150**.
(102) In the implementation depicted, the apparatus **102** may be fabricated in various ways. For example, the conductive elements **250** and insulators **308** may be arranged in layers during fabrication and then joined together in a single assembly. In another example, individual layers, such as with conductive elements **250** deposited thereon, may be fabricated, and then stacked to provide the single assembly.
(103) FIG. **4** illustrates at **400** a second arrangement of conductive elements **250** of the sensor elements **150** of the apparatus **102**, according to some implementations.
(104) As shown in the top view **402**, a plurality of sensor elements **150**(**1**)-(N) are distributed in a single row of columns across the apparatus **102**, forming an array. The apparatus control module **132** is located in a corner, and the conductors **270** connect the conductive elements **250** of the sensor elements **150** to the apparatus control module **132**.
(105) As described above, the arrangement of the apparatus control module **132**, the sensor elements **150** and their respective conductive elements **250**, and the conductors **270** may facilitate cutting the apparatus **102**.
(106) The side view **404** depicts an enlarged portion of a cross section of the apparatus **102** along line “B-B”. In the implementation shown, a cover layer **306** is provided. The cover layer **306** may comprise a dielectric material having a thickness of T**41**. In some implementations, the cover layer **306** may comprise a portion of the insulator **308**. The sensor elements **150** are arranged beneath the cover layer **306**.
(107) In this implementation, some sensor elements **150**, such as sensor element **150**(**1**), comprises three conductive elements **250**(**1**)-(**3**) that are stacked vertically with respect to one another.
(108) In comparison, some sensor elements **150**, such as sensor element **150**(**2**), comprise only two conductive elements **250**(**1**)-(**2**) that are stacked vertically with respect to one another.
(109) Each of the conductive elements **250**(**1**)-(**3**) have the same width W**41** and thickness T**42**. In one implementation, the width W**41** may be 20 mm. The conductive elements **250**(**1**)-(**3**) are arranged with the same spacing T**43** between the conductive elements **250** of the same sensor element **150**.
(110) An insulator **308** is disposed between the conductive elements **250**(**1**)-(**3**). Respective ones of the conductors **270** (not shown) are coupled to respective ones of the conductive elements **250**(**1**)-(**3**). A spacing width of W**42** is maintained between respective sensor elements **150**.
(111) In this implementation the apparatus **102** may comprise a first assembly **410** and a second assembly **412**. The first assembly **410** may comprise the cover layer **306**, insulator **308**(**1**), the first conductive element **250**(**1**), and the second conductive element **250**(**2**) for a plurality of sensor elements **150**. The second assembly **412** may comprise a third conductive element **250**(**3**) and associated insulator **308**(**2**).
(112) The first assembly **410** may be fabricated separately from the second assembly **412**. One or more instances of the second assembly **412** may then be applied to the first assembly **410**. In this fashion, some sensor elements **150** may comprise three conductive elements **250**(**1**)-(**3**), while other sensor elements **150** comprise two conductive elements **250**(**1**)-(**2**).
(113) As described above, the conductive elements **250** are electrically coupled to the apparatus control module **132** via the conductors **270** (not shown in this figure). In some implementations, the first assembly **410** may comprise the conductors **270**. One of the conductors **270** may terminate in a contact pad that is present at a surface of the first assembly **410**. For example, the contact pad may be located on the underside of the first assembly **410**. When the second assembly **412** is joined to the first assembly **410**, the contact pad may provide an electrically conductive pathway from the third conductive element **250**(**3**) to the conductors **270** in the first assembly **410**. For example, the contact may come into physical contact with an exposed portion of the third conductive element **250**(**3**).
(114) In some implementations the second assembly **412** may be encapsulated within an insulator **308**(**2**). A portion of the third conductive element

250(**3**) or an electrically conductive pathway thereto, may be configured to come into contact with the contact pad on the first assembly **410** when joined.
(115) FIG. **5** illustrates at **500** a third arrangement of conductive elements **250** of the sensor elements **150** of the apparatus **102**, according to some implementations.
(116) As shown in the top view **502**, a plurality of sensor elements **150**(**1**)-(N) are distributed in a single row of columns across the apparatus **102**, forming an array. The apparatus control module **132** is located in a corner, and the conductors **270** connect the conductive elements **250** of the sensor elements **150** to the apparatus control module **132**.
(117) As described above, the arrangement of the apparatus control module **132**, the sensor elements **150** and their respective conductive elements **250**, and the conductors **270** may facilitate cutting the apparatus **102**.
(118) The side view **504** depicts an enlarged portion of a cross section of the apparatus **102** along line “C-C”. In the implementation shown, a cover layer **306** is provided. The cover layer **306** has a thickness T**51**. The sensor elements **150** are arranged beneath the cover layer **306**.
(119) In this implementation, some sensor elements **150**, such as sensor element **150**(**1**), comprises three conductive elements **250**(**1**)-(**3**) that are stacked vertically with respect to one another.
(120) In comparison, some sensor elements **150**, such as sensor element **150**(**2**), comprise only two conductive elements **250**(**1**)-(**2**) that are stacked vertically with respect to one another.
(121) The conductive element **250**(**3**) has a width of W**53** and a thickness of T**52**. In one implementation, the width W**53** may be 20 mm. In comparison, the conductive elements **250**(**1**) and **250**(**2**) have a width of W**51**, where W**51** is less than W**53**.
(122) In one implementation, the conductive elements **250**(**2**)-**250**(**3**) may have a thickness of T**52**. In other implementations, different conductive elements **250** may have different thicknesses.
(123) The first conductive element **250**(**1**) and the second conductive element **250**(**2**) are separated by thickness T**53**. For example, T**53** may be between 1 and 3 millimeters (mm). In comparison, the conductive elements **250**(**2**) and **250**(**3**) are separated by thickness T**54**, where T**54** is less than T**53**. For example, T**54** may be less than 1 mm.
(124) An insulator **308** is disposed between the conductive elements **250**(**1**)-(**3**). Respective ones of the conductors **270** (not shown) are coupled to respective ones of the conductive elements **250**(**1**)-(**3**). A spacing width of W**52** is maintained between respective sensor elements **150**.
(125) In this implementation, the apparatus **102** may comprise a first assembly **510** and a second assembly **512**, as described above. For example, the first assembly **510** may comprise the cover layer **306**, insulator **308**(**1**), the first conductive element **250**(**1**), and the second conductive element **250**(**2**) for a plurality of sensor elements **150**. The second assembly **512** may comprise a third conductive element **250**(**3**) and associated insulator **308**(**2**).
(126) The first assembly **510** may be fabricated separately from the second assembly **512**. One or more instances of the second assembly **512** may then be applied to the first assembly **510**. In this fashion, some sensor elements **150** may comprise three conductive elements **250**(**1**)-(**3**), while other sensor elements **150** comprise two conductive elements **250**(**1**)-(**2**).
(127) As described above, the conductive elements **250** are electrically coupled to the apparatus control module **132** via the conductors **270** (not shown in this figure). In some implementations, the first assembly **510** may comprise the conductors **270**. One of the conductors **270** may terminate in a contact pad that is present at a surface of the first assembly **510**. For example, the contact pad may be located on the underside of the first assembly **510**. When the second assembly **512** is joined to the first assembly **510**, the contact pad may provide an electrically conductive pathway from the third conductive element **250**(**3**) to the conductors **270** in the first assembly **510**. For example, the contact may come into physical contact with an exposed portion of the third conductive element **250**(**3**).
(128) In some implementations the second assembly **512** may be encapsulated within an insulator **308**(**2**). A portion of the third conductive element **250**(**3**) or an electrically conductive pathway thereto, may be configured to come into contact with the contact pad on the first assembly **510** when joined.
(129) FIG. **6** illustrates at **600** a fourth arrangement of conductive elements **250** of the sensor elements **150** of the apparatus **102**, according to some implementations.
(130) As shown in the top view **602**, a plurality of sensor elements **150**(**1**)-(N) are distributed in a single row of columns across the apparatus **102**, forming an array. The apparatus control module **132** is located in a corner, and the conductors **270** connect the conductive elements **250** of the sensor elements **150** to the apparatus control module **132**.
(131) As described above, the arrangement of the apparatus control module **132**, the sensor elements **150** and their respective conductive elements **250**, and the conductors **270** may facilitate cutting the apparatus **102**.
(132) The side view **604** depicts an enlarged portion of a cross section of the apparatus **102** along line “D-D”. In the implementation shown, a cover layer **306** is provided. The sensor elements **150** are arranged beneath the cover layer **306**. In some implementations, the cover layer **306** may comprise a portion of the insulator **308**.
(133) In this implementation, each sensor element **150** comprises a first conductive element **250**(**1**) and a second conductive elements **250**(**2**) stacked vertically with respect to one another. The third conductive element **250**(**3**) is coplanar with the first conductive element **250**(**1**). For example, the first conductive element **250**(**1**) and the third conductive element **250**(**3**) may both be arranged just beneath the cover layer **306**, may be deposited on the same substrate, and so forth.
(134) The conductive elements **250**(**1**) and **250**(**2**) have a width of W**61**. In one implementation, the width W**61** may be 20 mm. In comparison, the conductive element **250**(**3**) has a width of W**62**, where W**62** is less than W**61**.
(135) An insulator **308** is disposed between the conductive elements **250**(**1**)-(**3**). Respective ones of the conductors **270** (not shown) are coupled to respective ones of the conductive elements **250**(**1**)-(**3**).
(136) In one implementation, during operation a sensor element **150** may comprise the first conductive element **250**(**1**), the second conductive element **250**(**2**), and one or more of the third conductive elements **250**(**3**) that are adjacent to the first conductive element **250**(**1**).
(137) FIG. **7** illustrates at **700** operation of the apparatus **102** with an item management accessory **702**, according to some implementations. The various configurations of conductive elements **250**(**1**)-(**4**) described with regard to FIGS. **3**A-**6** may be used in conjunction with the cover layer **306**. As mentioned above, respective conductive elements **250** may comprise expanded conductive elements **310**. In some implementations, compensation sensor elements **372** (not shown), may be included.
(138) An item management accessory **702** may provide a gap **704** between an upper surface of the cover layer **306** and the items **106** being stowed. The item management accessory **702** may be designed to maintain an orderly arrangement of items **106**, providing physical dividers or walls between lanes **104**. In some implementations the item management accessory **702** may include pushers or other mechanisms designed to move items **106** towards a front of the inventory location. In another implementation the item management accessory **702** may provide airflow to facilitate temperature or humidity control of the items **106**.
(139) The operation of the sensor elements **150** by the apparatus control module **132** in the self-capacitance mode allows for the items **106** to be detected even with the presence of the gap **704**. In some implementations, the gap **704** may be at least 0.1 mm. For example, the gap **704** may be between 0.1 mm and 8 mm. In some implementations, the item management accessory **702** may be used to provide the gap **704** to improve performance of the system **100**, as described with regard to FIG. **8**.
(140) FIG. **8** illustrates at **800** the apparatus **102** with an extended cover layer **802**, according to some implementations. The various configurations of conductive elements **250**(**1**)-(**4**) described with regard to FIGS. **3**A-**6** may be used in conjunction with the extended cover layer **802**. As mentioned

above, respective conductive elements **250** may comprise expanded conductive elements **310**. In some implementations, compensation sensor elements **372** (not shown), may be included.
(141) The extended cover layer **802** may comprise a dielectric material, such as non-conductive plastic, ceramic, and so forth. The sensor elements **150** are arranged beneath the extended cover layer **802**. In some implementations, the extended cover layer **802** may be placed over the cover layer **306**.
(142) The extended cover layer **802** has a thickness T**81**. In some implementations, the thickness T**81** may be at least 0.1 mm. For example, the thickness T**81** may be between 0.1 mm and 8 mm. For example, T**81** may be at least 3 mm. This increases the distance between the conductive elements **250** and the item **106** during use, similar to the gap **704** provided by the item management accessory **702**. A useful result of increasing the thickness T**81** is improved resistance to adverse effects caused by the presence of fluids, such as water, on the capacitance measurements. For example, a thickness T**81** of 5 mm provides reliable operation in the presence of large droplets and small pools of water.
(143) A graph **804** shows the relationship between distance **810** and capacitance **812**. Capacitance **812** varies with distance **810** in a 1/x hyperbolic relationship. The thickness T**81** is shown with respect to the graph **804**. By introducing the spacing produced by thickness T**81** of the extended cover layer **802**, or the gap **704** of the item management accessory **702**, the system will be assessing capacitance values that exhibit more of a linear relationship, and less of an exponential relationship. Colloquially, the portion of the curve being assessed is “less steep”. This provides a substantial advantage during operation, particularly in situations where the items **106** have non-uniform lower portions or may not sit flat. For example, some plastic beverage bottles have an arrangement of protrusions, providing a non-uniform surface to the apparatus **102**. In another example, items **106** may not sit completely vertically, either due to irregularities in their containers, displacement during pick or place operations, or other factors.
(144) As a result of the spacing introduced by the extended cover layer **802** or the item management accessory **702**, the irregularities of the items **106** result in less of a capacitance change than if the items **106** were closer to the conductive elements **250**. As a result, during operation the resulting capacitance data **116** contains less noise, and results in simpler data processing to determine the interaction data **128**.
(145) FIG. **9** illustrates at **900** several layouts of sensor elements **150** of the apparatus **102**, according to some implementations.
(146) A first layout **902** depicts the sensor elements **150** arranged in pairs, with each pair comprising a first triangular sensor element **150** and a second triangular sensor element **150**. Each of the triangular sensor elements **150** has three sides forming the triangular shape. Each of the triangular sensor elements **150** may be a right triangle. The longest side of each triangle, or hypotenuse, may be arranged to be adjacent to the longest side of the other triangle in the pair. When arranged in this manner, an overall shape of the pair of triangular sensor elements **150** is generally square or rectangular. With regard to the first layout **902**, a long axis through the pair of triangular sensor elements **150** extends from a front to a back of the apparatus **102**. Operation of the first layout **902** is discussed in more detail with regard to FIG. **10**.
(147) A second layout **904** depicts an arrangement in which the sensor elements **150** are triangular sensor elements **150** arranged adjacent to one another. In the arrangement depicted, the triangles are isosceles triangles. In other implementations, other types of triangles or other geometric shapes may be used.
(148) A third layout **906** depicts an arrangement in which the sensor elements **150** comprise strips or rectangles having a long axis that extends from the front to the back of the apparatus **102** or other inventory location. In some implementations, the width of the strips may be configured to be less than or equal to the width of a lane **104**.
(149) A fourth layout **908** depicts an arrangement in which the sensor elements **150** comprise strips or rectangles having a long axis that extends from the front to the back of the apparatus **102** or other inventory location, with additional sensor elements **150** arranged at one or both ends of each strip. Operation of this layout is described in more detail with regard to FIG. **23**.
(150) A fifth layout **910** depicts the sensor elements **150** arranged in pairs, with each pair comprising a first sensor element **150** and a second sensor element **150** arranged in a “V” shape. In this layout, the apex of each “V” is on the same side of the apparatus **102**.
(151) A sixth layout **912** depicts the sensor elements **150** arranged in pairs, with each pair comprising a first sensor element **150** and a second sensor element **150** arranged in a “V” shape. In this layout, the apex of each “V” is on an opposite side of the apparatus **102** relative to an adjacent pair of sensor elements **150**.
(152) A seventh layout **914** depicts an arrangement in which the sensor elements **150** comprise strips or rectangles having a long axis that extends from the front to the back of the apparatus **102** or other inventory location, and one or more compensation sensor elements **372** are present. In some implementations, the width of the strips of the sensor elements **150** may be configured to be less than or equal to the width of a lane **104**.
(153) In the implementation shown, three compensation sensor elements **372**(**1**)-(**3**) are present. The compensation sensor elements **372** are distributed to the left, center, and right, of the array of sensor elements **150**(**1**)-(N). In other implementations the compensation sensor element(s) **372** may be placed in other locations with respect to the apparatus **102**.
(154) In some implementations, layouts **902**, **904**, **910**, and **912** may be operated as described with regard to FIG. **10**.
(155) In other implementations, other layouts may be used. For example, a grid layout may be used, comprising sensor elements **150** arranged in rows and columns.
(156) FIG. **10** illustrates at **1000** operation of triangular sensor elements **150** of the apparatus **102**, according to some implementations. As described above with regard to FIG. **9**, pairs of sensor elements **150** may be operated to determine additional information about the items **106** proximate to the apparatus **102**. For ease of discussion with regard to this figure, it is assumed that each lane **104** includes a single pair of sensor elements **150**(**1**)-(**2**). It is understood that in other implementations, other arrangements with respect to lanes **104** may be used.
(157) An item location configuration **1002** is depicted in which sensor element **150**(**1**) is an active element **1006**. An active element **1006** may comprise one or more sensor elements **150** that are connected via the switch module **212** to the capacitive sensor module **206** to determine capacitance data **116** at the same time. For example, the apparatus control module **132** is used to determine capacitance data **116** associated with the sensor element **150**(**1**). Due to the change in area of the sensor element **150**(**1**) with respect to the depth of the lane **104**, a change in the capacitance data **116** may be associated with a depth along the sensor element **150**(**1**). In some implementations, the apparatus control module **132** may determine individual capacitance data **116** for each of the sensor elements **150**(**1**)-(**2**), and use the pair of capacitance data **116** to determine the location.
(158) An item count configuration **1004** is shown in which both sensor elements **150**(**1**)-(**2**) are operated simultaneously as active elements **1006**. For example, the switch module **212** may connect the first conductive element **250**(**1**)(**1**) of the first sensor element **150**(**1**) and the first conductive element **250**(**2**)(**1**) of the second sensor element **150**(**2**) at the same time to the signal source of the capacitive sensor module **206**.
(159) FIG. **11** is a graph **1100** of capacitance over time depicting various interactions as determined by the apparatus **102**, according to some implementations. These interactions produce well defined changes in capacitance that may be used to determine one or more attributes of the interaction data **128**.
(160) A horizontal axis depicts time **1102** in seconds, increasing from left to right. A vertical axis depicts capacitance **1104** as determined by the apparatus control module **132**.
(161) At time 0, a first item **106**(**1**) is present at the apparatus **102** and is detected.
(162) At **1120** the first item **106**(**1**) is removed. The momentary increase in capacitance may be due to the additional capacitance of the user **108**, such as their hand, while the overall decrease in capacitance is due to the removal of the item **106**(**1**). The system **100** may use this information to determine the presence of a user **108**.
(163) At **1122** the user **108** is holding the first item **106**(**1**) approaches the apparatus **102** and places the first item **106**(**1**) at the apparatus **102**, resulting in an increase in capacitance.
(164) At **1124** the user **108** removes their hand, leaving the first item **106**(**1**) at the apparatus **102**, resulting in a nonzero capacitance.

(165) At **1126** the user **108** is holding a second item **106**(**2**) and approaches the apparatus **102** and places the second item **106**(**2**) at the apparatus **102**. At **1128** the user **108** removes their hand, leaving the second item **106**(**2**) and the first time **106**(**1**) at the apparatus **102**, resulting in an increase in the capacitance.
(166) FIG. **12** depicts a block diagram **1200** of a first implementation to determine interaction data **128**. The apparatus **102** includes the apparatus control module **132** that provides as output the capacitance data **116**. The capacitance data **116** may be sent to the analysis module **124** using the communication interface **260**. For example, the apparatus control module **132** may use a wireless communication interface or wired communication interface to send the capacitance data **116** to the analysis module **124**. Language indicative of a particular time may indicate a particular instant in time or a particular interval of time that begins or ends at the designated particular time.
(167) The analysis module **124** may comprise instructions that execute on one or more computing devices. For example, the one or more computing devices may be located in the same physical location as the apparatus **102**, or at a remote location such as at a cloud data center. The analysis module **124** may include an event detection module **1202**. The event detection model **1202** may accept as input one or more of capacitance data **116**, item data **126**, or other data **114**. The event detection module **1202** is configured to generate event data **1204**. The event data **1204** may comprise information indicative of a change deemed to be significant that is associated with an inventory location or portion thereof. For example, the event data **1204** may be indicative of a determination by a set of one or more sensor elements **150** of proximity of an object, such as an item **106** or user **108**. In another example, the event data **1204** may comprise an indication that a weight change has exceeded a threshold value. In yet another example, the event data **1204** may indicate that motion between a plurality of images has been detected that exceeds a threshold value. The event detection module **1202** may utilize one or more filter functions, comparison functions, and so forth, to determine the event data **1204**. For example, the event data **1204** may result from a determination that the capacitance data **116** and other data **114** have each experienced changes that exceed a threshold value. The event detection module **1202** may utilize various rules or conditions to determine the occurrence of an event and subsequent generation of event data **1204**.
(168) During operation, the analysis module **124** may sum the capacitance data **116** acquired from two or more sensor elements **150**. For example, first capacitance data **116**(**1**) acquired using sensor element **150**(**14**) and second capacitance data **116**(**2**) acquired using sensor element **150**(**15**) may be combined via summing, averaging, or other techniques to determine information about lane **104**(**7**), such as shown in FIG. **17**.
(169) The analysis module **124** may accept as input event data **1204**, capacitance data **116**, or other data **114** and perform one or more data processing functions. The data processing functions may include, but are not limited to, filtering, noise reduction, signal recovery, determination of a moving average, statistical analysis, and so forth. In one implementation, the event data **1204** may be utilized to apply particular data processing functions to specific data or portions thereof. For example, the event data **1204** may indicate presence of an object based on a variation in the capacitance data **116**. The event data **1204** may be associated with a particular time. The analysis module **124** may access one or more of a set of capacitance data **116** or the other data **114** that was obtained before and after the particular time. The analysis module **124** may apply one or more noise filters or other data processing functions to determine baseline data before the particular time of the event data **1204** and after the particular time.
(170) The analysis module **124** may also accept as input other data **114**, such as input from other sensor(s) **112** a weight sensor, light sensor, accelerometer, hygrometer, radio frequency (RF) receiver, and so forth. The analysis module **124** may process the other data **114** to determine the interaction data **128**. For example, image data may be processed to generate information indicative of changes between images, object recognition data, and so forth. Processing of one or more of the image data or portions thereof may be performed by implementing, at least in part, one or more of the following tools or techniques. In one implementation, processing of the image data may be performed, at least in part, using one or more tools available in the OpenCV library as developed by Intel Corporation of Santa Clara, California, USA; Willow Garage of Menlo Park, California, USA; and Itseez of Nizhny Novgorod, Russia. In another implementation, functions available in the OKAO machine vision library as promulgated by Omron Corporation of Kyoto, Japan, may be used to process the image data. In still another implementation, functions such as those in the Machine Vision Toolbox for Matlab (MVTB) available using MATLAB as developed by MathWorks, Inc. of Natick, Massachusetts, USA, may be utilized.
(171) An interaction data module **1210** may be configured to use as input one or more of the event data **1204**, item data **126**, capacitance data **116**, other data **114**, and so forth, to generate the interaction data **128**. The interaction data module **1210** may include a location estimation module **1212**. The location estimation module **1212** may be configured to use one or more of the lane configuration data **130**, the event data **1204**, the capacitance data **116**, or the other data **114** to determine location data **1214** associated with an interaction.
(172) The location estimation module **1212** may also use the item data **126** to determine the location data **1214**. For example, the item data **126** associated with a particular type of item **106** may be retrieved, providing information such as a width and depth, measured capacitance of a representative sample of the type of item **106**, and so forth. The location estimation module **1212** may use information such as the width and depth to determine an area that item **106** is expected to cover. Based on this information, the location estimation module **1212** may avoid generating location data **1214** that places the item **106** in the middle of a divider or some other unlikely location. For example, an item area associated with the item **106** may be determined by multiplying the width and depth indicated by the item data **126**. An inventory area designated for stowage of items **106** is determined that is associated with the inventory location. For example, lane **104**(**1**) may be indicated in physical layout data as having a particular location in the facility and a particular width and depth. The location estimation module **1212** may determine location data **1214** indicative of a center of the item area for the type of item **106** such that the item area is arranged within the inventory area. Continuing the example, the location estimation module **1212** may be configured to avoid generating location data **1214** that would place an item halfway off the apparatus **102**, or spanning a divider between lanes **104**.
(173) The location data **1214** may provide information indicative of a particular location with regard to an inventory location. For example, the location data **1214** may indicate a particular lane **104**, portion of a lane **104**, coordinates of the point or area on the apparatus **102**, and so forth. The interaction may comprise movement, presence, pick, place, and so forth, of an object or an item **106**.
(174) In one implementation, the event data **1204** may specify a particular lane **104** at which an event took place, whether the type of event was a pick or a place, and a change in capacitance before and after the event. In this implementation, the lane **104** may be used as the location data **1214**.
(175) In another implementation, the event data **1204** may provide information associated with or sets of one or more sensor elements **150**. The lane configuration data **130** may then be used to determine which lanes **104** are associated with the sets of sensor elements **150**.
(176) In one implementation, the location estimation module **1212** may use the capacitance data **116** obtained from one or more sets of sensor elements **150** on the apparatus **102** to determine that there was a change made to the items **106** at a given lane **104**. For example, the estimated location may correspond to an area of a set of sensor elements **150** used to determine the capacitance data **116**. In another example, the capacitance data **116** from a plurality of sensor elements **150** in a known configuration, such as depicted in layouts **902**, **904**, **910**, or **912**, may be used to generate the location data **1214**. The interaction data module **1210** may access the item data **126** to determine the type of item **106** stored at the lane **104**(**1**), and other characteristics about that type of item **106** such as a per item weight.
(177) The interaction data module **1210** may use other modules (not shown) to determine a quantity. For example, weight data from weight sensors at the apparatus **102** may be used to determine a weight change value, and this weight change value may be compared with the per item weight to determine a quantity of items **106** that have changed. A sign of the weight change may be used to indicate addition or removal of an item **106**. For example, a positive weight change value may indicate an increase in weight indicative of placement of an item **106** at an inventory location, while a negative weight change value may indicate a decrease in the weight indicative of pick of an item **106** from the inventory location. The interaction data module **1210** may thus generate interaction data **128**. For example, the interaction data **128** may indicate a quantity and type of item **106** that was removed from a particular inventory location.
(178) FIG. **13** depicts a block diagram **1300** of a second implementation to determine interaction data **128**. In this implementation, the apparatus **102**

includes the apparatus control module **132** and the event detection module **1202**. During operation, the apparatus control module **132** sends capacitance data **116** to the event detection module **1202**. The event detection module **1202** may utilize the lane configuration data **130** to determine the event data **1204**. The event data **1204** may then be sent to the analysis module **124** executing on another computing device. The analysis module **124** may then process the event data **1204** as described above to determine the interaction data **128**.
(179) During operation, the apparatus control module **132** may sum the capacitance data **116** acquired from two or more sensor elements **150**. For example, first capacitance data **116**(**1**) acquired using sensor element **150**(**14**) and second capacitance data **116**(**2**) acquired using sensor element **150**(**15**) may be combined via summing, averaging, or other techniques to determine capacitance data **116** that is associated with lane **104**(**7**), such as shown in FIG. **17**.
(180) The event data **1204** provided by the event detection module **1202** of the apparatus **102** may comprise information indicative of an estimated location, such as a lane **104**, a type of interaction such as a pick or place, and may include information such as a quantity of items **106** added or removed, and so forth. For example, the determination of a pick or place may be based on a difference in capacitance values before and after an event. The mathematical sign of the difference between those values may be used to determine a pick or a place of an item **106**.
(181) The event detection module **1202** or other modules such as a portion of the analysis module **124** may utilize various methods to determine one or more of the event data **1204** or the interaction data **128**. An interrupt method is discussed with regard to FIG. **14**. A polling method is discussed with regard to FIG. **15**.
(182) FIG. **14** depicts at **1400** graphs of capacitance and an interrupt-based technique to accumulate item count over time responsive to various interactions, according to some implementations. For ease of illustration, and not as a limitation, these graphs are associated with a single lane **104** used to stow homogenous items **106**.
(183) As described with regard to FIG. **11**, several interactions produce well defined changes in capacitance that may be used to determine one or more attributes of the event data **1204** or the interaction data **128**. A first graph **1402** depicts changes in capacitance over time while a second graph **1404** depicts accumulated item count.
(184) A horizontal axis depicts time **1410** in seconds, increasing from left to right. With regard to the first graph **1402**, a vertical axis depicts capacitance **1412** as determined by the apparatus control module **132**. With regard to the second graph **1404**, a vertical axis depicts an accumulated item count **1414**.
(185) In this illustration, a threshold value **1450** of capacitance is specified. The threshold value **1450** may be determined based on testing, item data **126**, during initial setup of the apparatus **102**, and so forth. With regard to the interrupt method depicted, a change in capacitance **1412** that exceeds the threshold **1450** generates an interrupt that updates the value of the accumulated item count **1414**. The mathematical sign of the difference in the capacitance may be used to determine if the interaction is a pick or a place. For example, a positive change in capacitance may indicate a place, while a negative change in capacitance may indicate a pick. In some implementations the magnitude of the change may be indicative of the quantity of items **106** picked or placed.
(186) The technique may begin operation with a known quantity of items **106** at the apparatus **102**. For example, at time 0 no items **106** are present. Six events **1420** through **1430** are depicted.
(187) At **1420** a positive change in capacitance is noted that exceeds the threshold **1450**, with an amplitude indicative of one item **106** being placed. The accumulated item count **1414** is correspondingly updated to a value of “1”.
(188) At **1422** two items **106** are placed. The positive change in capacitance is noted that exceeds the threshold **1450**, with an amplitude indicative of two items **106**, corresponding to two items **106** being placed. The accumulated item count **1414** is correspondingly updated to a value of “3”.
(189) At **1424** a positive change in capacitance is noted that exceeds the threshold **1450**, with an amplitude indicative of one item **106** being placed. The accumulated item count **1414** is correspondingly updated to a value of “4”.
(190) At **1426** a negative change in capacitance is noted that exceeds the threshold **1450**, with an amplitude indicative of one item **106** being picked. The accumulated item count **1414** is correspondingly updated to a value of “3”.
(191) At **1428** a positive change in capacitance is noted that exceeds the threshold **1450**, with an amplitude indicative of one item **106** being placed. The accumulated item count **1414** is correspondingly updated to a value of “4”.
(192) At **1430** a negative change in capacitance is noted that exceeds the threshold **1450**, with an amplitude indicative of two items **106** being picked. The accumulated item count **1414** is correspondingly updated to a value of “2”.
(193) Between the events, the apparatus **102** may be operated in a low-power mode. For example, as described below, the apparatus **102** may be operated in a proximity detection mode that determines capacitance data **116** at a low sample rate, such as once every 500 milliseconds, and for a specified set of sensor elements **150**, such as all. Upon determining the presence of an object, such as the user **108**, the apparatus **102** may transition from a low power mode to an active mode during which capacitance data **116** is determined at a faster sample rate such as every 100 ms and for sets of sensor elements **150** that are associated with particular lanes **104**.
(194) FIG. **15** depicts at **1500** graphs of capacitance and a polling-based technique to accumulate item count over time responsive to various interactions, according to some implementations. For ease of illustration, and not as a limitation, these graphs are associated with a single lane **104** used to stow homogenous items **106**. In this illustration, at time 0 the lane **104** is empty.
(195) As described with regard to FIG. **11**, several interactions produce well defined changes in capacitance that may be used to determine one or more attributes of the event data **1204** or the interaction data **128**. A first graph **1502** depicts changes in capacitance over time while a second graph **1504** depicts an accumulated item count.
(196) A horizontal axis depicts time **1510** in seconds, increasing from left to right. With regard to the first graph **1502**, a vertical axis depicts capacitance **1512** as determined by the apparatus control module **132**. With regard to the second graph **1504**, a vertical axis depicts an accumulated item count **1514**.
(197) Shown in these graphs are five individual instances of one item placed **1520** and five individual instances of one item picked **1522**. During operation, the apparatus control module **132** performs an ongoing polling of the set of sensor elements **150** associated with the lane **104**, providing ongoing capacitance **1512** values at successive times.
(198) As items **106** are placed, the capacitance **1512** increases. As items **106** are picked, the capacitance **1512** decreases. After an event takes place, such as a pick or a place, a dynamic baseline **1550** is determined. For example, the dynamic baseline **1550** may comprise a simple average, moving average, linear fit, and so forth of the capacitance values for a specified interval of time. A threshold value **1552** is specified, relative to the dynamic baseline **1550**. The threshold value **1550** may be determined based on testing, item data **126**, during initial setup of the apparatus **102**, and so forth.
(199) With regard to the polling method depicted, a change in capacitance **1512** with respect to the dynamic baseline that exceeds the threshold **1552** may be used to update the value of the accumulated item count **1514**. The mathematical sign of the difference in the capacitance relative to the dynamic baseline **1550** may be used to determine if the interaction is a pick or a place. For example, a positive change in capacitance may indicate a place, while a negative change in capacitance may indicate a pick. In some implementations the magnitude of the change may be indicative of the quantity of items **106** picked or placed.
(200) With regard to the illustration, as items **106** are successively added, the capacitance **1512** increases. The dynamic baseline **1550**, and associated threshold **1552** are also increased. As items **106** are successively picked, the capacitance **1512** decreases. The dynamic baseline **1550**, and associated threshold **1552** are also decreased.
(201) In some implementations, the apparatus **102** may be operated using the polling-based technique to determine and maintain capacitance data **116** over an extended period of time. The data from the extended period of time may be used to characterize operation of the apparatus **102**. For example,

drift in the capacitance **1512** over time may be characterized using the data obtained using the polling-based technique.
(202) In some circumstances both the interrupt-based technique and the polling-based technique may be used contemporaneously by the same apparatus **102**.
(203) FIG. **16** is a flow diagram **1600** of a process for determining the association between sensor elements **150** and lanes **104** at an inventory location, according to some implementations. The process may be implemented at least in part by the apparatus control module **132**. In some situations, it may be advantageous to determine the association of sets of one or more sensor elements **150** and a particular lane **104**. This association may be stored as the lane configuration data **130**.
(204) At **1602** a first lane **104**(**1**) and a second lane **104**(**2**) associated with an apparatus **102** are determined. The apparatus **102** comprises a first set of sensors elements **150** and a second set of sensor elements **150**. For example, the first set of sensor elements **150** may comprise the first sensor element **150**(**1**) and the second set of sensor elements **150** may comprise the second sensor element **150**(**2**).
(205) At **1604**, using the first set of sensor elements, first capacitance data **116** is determined.
(206) At **1606**, using the second set of sensor elements, second capacitance data **116** is determined.
(207) At **1608** a first item **106** is placed at the first lane **104**(**1**). A determination as to the placement of the first item **106** may be based on one or more of a difference between the first capacitance data **116** and the second capacitance data **116**, user input, expiration of a timer after presentation of output using an output device such as a light or speaker, and so forth. For example, during this process a user interface may be presented on a tablet computer that prompts to user to "place one item in the first lane" and includes a user input control that allows indication that this operation has been completed.
(208) In another implementation, before **1604**, a first item **106** may be placed at the first lane **104**(**1**) and at **1608** the first item **106** may be removed.
(209) At **1610**, using the first set of sensor elements, third capacitance data **116** is determined.
(210) At **1612**, using the second set of sensor elements, fourth capacitance data **116** is determined.
(211) At **1614** a first difference is determined between the first capacitance data **116** and the third capacitance data **116**.
(212) At **1616** a second difference is determined between the second capacitance data **116** and the fourth capacitance data **116**.
(213) Based on the first difference and the second difference, an association between the first lane **104**(**1**) and one or more of the first set of sensor elements **150** or the second set of sensor elements **150** may be made. One implementation of this association is depicted with regard to the determinations of **1618** and **1624**.
(214) At **1618** if the first difference is greater than a threshold value, at **1620** the first set of sensor elements **150** are determined to be associated with the first lane **104**(**1**). If not, at **1622** the first set of sensor elements **150** are determined to not be associated with the first lane **104**(**1**).
(215) At **1624** if the second difference is greater than a threshold value, at **1626** the second set of sensor elements **150** are determined to be associated with the first lane **104**(**1**). If not, at **1628** the second set of sensor elements **150** are determined to not be associated with the first lane **104**(**1**).
(216) The process may be iterated, using various combinations of sets of sensor elements **150**.
(217) Once a set of sensor elements **150** have been associated with a particular lane **104**, during operation the system **100** may combine the data from the individual sensor elements **150** in the set. For example, if the set of sensor elements **150** comprises a first sensor element **150**(**1**) and a second sensor element **150**(**2**), the apparatus control module **132** or another computing device may sum the capacitance data **116** associated with the two to determine the capacitance data **116** for the set of sensor elements **150**.
(218) FIG. **17** depicts at **1700** active and inactive sensor elements **150** associated with respective lanes **104**, according to some implementations. During operation, the apparatus control module **132** may selectively connect particular sensor elements **150** using the switch module **212**.
(219) A first view **1702** depicts a top view of the apparatus **102** showing rectangular sensor elements **150**(**1**)-(**20**) and lanes **104**(**1**)-(**8**) with items **106**. In some implementations, such as shown here, the relative arrangement of lanes **104** to sensor elements **150** may not be one-to-one. For example, some items **106** and their associated lanes **104** are wider than other items **106**. Also, the left-to-right edges of lanes **104** may differ from the left-to-right edges of the individual sensor elements **150**. The ability to support these various configurations significantly improves the usability of the system **100**.
(220) A second view **1704** depicts the top view of the apparatus **102** with the items **106** removed. The second view **1704** depicts inactive elements **1706** and active elements **1708**. While determining the capacitance data **116**, inactive elements **1706** may comprise those sensor elements **150** that are omitted from use by the apparatus control module **132**. For example, during operation the switch module **212** may not connect the inactive elements **1706** to the signal source of the capacitive sensor module **206**. In comparison, an active element **1708** may comprise one or more sensor elements **150** that are connected via the switch module **212** to the capacitive sensor module **206** to determine capacitance data **116** at the same time. For example, the apparatus control module **132** is used to determine capacitance data **116** associated with the sensor element **150**(**1**).
(221) By selectively operating the sensor elements **150**, the quality of the interaction data **128** may be improved. For example, with regard to lane **104**(**3**), sensor element **150**(**6**) is an active element **1708** while immediately adjacent sensor elements **150**(**5**) and **150**(**7**) are inactive elements **1706**. This results in capacitance data **116** that is better representative of the changes associated with lane **104**(**3**), and less prone to changes due to interactions with adjacent lanes **104**(**2**) or **104**(**4**).
(222) By selectively operating the sensor elements **150**, other operational factors may also be improved. For example, by not using the inactive elements **1706**, the quantity of capacitance data **116** generated and subsequently processed is reduced. In another example, by not using the inactive elements **1706**, power consumption and corresponding power dissipation is reduced. In yet another example, by not using the inactive elements **1706**, system latency is reduced as only the active elements **1708** are scanned.
(223) FIG. **18**A depicts at **1800** block diagrams of the conductive elements **250** and respective connections for a first mode and a second mode, according to some implementations. The operations associated with these modes may be implemented at least in part by the apparatus control module **132**, or portions thereof as shown. These modes may be utilized with various configurations of conductive elements **250**, such as described herein.
(224) A first mode **1802** depicts the configuration of the conductive elements **250** during capacitance sensing. For example, the first mode **1802** may be used to determine the load capacitance data **254**.
(225) While in the first mode **1802**, the switch module **212** connects a sensor terminal of the capacitive sensor module **206**, such as connected to a capacitive signal source, to the first conductive element **250**(**1**). The first conductive element **250**(**1**) is beneath the cover layer **306**.
(226) While in the first mode **1802**, the switch module **212** connects a shield terminal of the capacitive sensor module **206**, such as connected to a shield signal source, to the second conductive element **250**(**2**). The second conductive element **250**(**2**) is located beneath the first conductive element **250**(**1**).
(227) While in the first mode **1802**, the switch module **212** connects the third conductive element **250**(**3**) to the ground **248**. The capacitive sensor module **206** may also be connected to the ground **248**. In the first mode **1802**, during operation the relative arrangement of the conductive elements **250**(**1**)-(**3**) an electric field extends above the cover layer **306**.
(228) A second mode **1804** depicts the configuration of the conductive elements **250** during compensation sensing. For example, the second mode **1804** may be used to determine the reference capacitance data **256**. Due to various factors, such as variations in manufacture, ambient temperature, temperature of an item **106** that is in contact with the apparatus **102**, and so forth, the capacitance values determined may vary. The second mode **1804** may be used to compensate for these variations.
(229) While in the second mode **1804**, the switch module **212** connects the first conductive element **250**(**1**) and the third conductive element **250**(**3**) to the ground **248**.

(230) While in the second mode **1804**, the switch module **212** connects the second conductive element **250**(**2**) to the capacitive signal source.
(231) In the second mode **1804**, during operation the relative arrangement of the conductive elements **250**(**1**)-(**3**) and an electric field extends between the conductive elements **250**(**1**)-(**3**) and without substantial extension above the cover layer **306**. The resulting reference capacitance data **256** is representative of the capacitance between the conductive elements **250**(**1**)-(**3**), and not due to a load on the apparatus **102** or an underlying supporting shelf or other structure.
(232) FIG. **18**B depicts at **1850** block diagrams of the conductive elements **250** and respective connections for a first variation and a second variation of the second mode, according to some implementations. The operations associated with this mode and variations may be implemented at least in part by the apparatus control module **132**, or portions thereof as shown. These modes and variations may be utilized with various configurations of conductive elements **250**, such as described herein.
(233) At **1852** a first variation of the second mode **1804** is shown that depicts the configuration of the conductive elements **250** of a compensation sensor element **372** during compensation sensing. For example, the second mode **1804** may be used to determine the reference capacitance data **256**. Due to various factors, such as variations in manufacture, ambient temperature, temperature of an item **106** that is in contact with the apparatus **102**, and so forth, the capacitance values determined may vary. The second mode **1804** may be used to compensate for these variations.
(234) While in the first variation of the second mode **1804**, the switch module **212** connects the first conductive element **250**(**1**) to the ground **248**.
(235) While in the first variation of the second mode **1804**, the switch module **212** connects the second conductive element **250**(**2**) and the fourth conductive element **250**(**4**) to the shield signal source.
(236) While in the first variation of the second mode **1804**, the switch module **212** connects the third conductive element **250**(**3**) to the signal source.
(237) The resulting reference capacitance data **256** is representative of the capacitance between the conductive elements **250**(**1**)-(**4**), and not due to a load on the apparatus **102** or an underlying supporting shelf or other structure.
(238) At **1854** a second variation of the second mode **1804** is shown that depicts the configuration of the conductive elements **250** of a compensation sensor element **372** during compensation sensing. For example, the second mode **1804** may be used to determine the reference capacitance data **256**. Due to various factors, such as variations in manufacture, ambient temperature, temperature of an item **106** that is in contact with the apparatus **102**, and so forth, the capacitance values determined may vary. The second mode **1804** may be used to compensate for these variations.
(239) While in the second variation of the second mode **1804**, the switch module **212** connects the first conductive element **250**(**1**) and the fourth conductive elements **250**(**4**) to the ground **248**.
(240) While in the second variation of the second mode **1804**, the switch module **212** connects the second conductive element **250**(**2**) to the shield signal source.
(241) While in the second variation of the second mode **1804**, the switch module **212** connects the third conductive element **250**(**3**) to the signal source.
(242) The resulting reference capacitance data **256** is representative of the capacitance between the conductive elements **250**(**1**)-(**4**), and not due to a load on the apparatus **102** or an underlying supporting shelf or other structure.
(243) FIG. **19**A depicts at **1900** conductive element configurations for a first conductive element arrangement in the first mode **1802** and the second mode **1804**, according to some implementations. The configurations associated with these modes may be implemented at least in part by the apparatus control module **132**, or portions thereof as shown.
(244) The configuration depicted comprises conductive elements **250** arranged as described with respect to FIG. **4**. The sensor element **150**(**1**) has three conductive elements **250**(**1**)-(**3**) while other sensor elements **150**(**2**)-(**3**) have two conductive elements **250**(**1**)-(**2**). As described above, in some implementations the third conductive elements **250**(**3**) may comprise a second assembly.
(245) In the first mode **1802** while operating the first sensor element **150**(**1**), the switch module **212** connects the conductive elements **250**(**1**)-(**3**) as discussed with regard to FIG. **18**.
(246) While in the first mode **1802** and operating the second sensor element **150**(**2**), the switch module **212** connects a sensor terminal of the capacitive sensor module **206**, such as connected to a capacitive signal source, to the first conductive element **250**(**1**) of the second sensor element **150**(**2**). The switch module **212** also connects a shield terminal of the capacitive sensor module **206**, such as connected to a shield signal source, to the second conductive element **250**(**2**) of the second sensor element **150**(**2**). The second conductive element **250**(**2**) of the second sensor element **150**(**2**) is located beneath the first conductive element **250**(**1**) of the second sensor element **150**(**2**).
(247) While in the first mode **1802** and operating the second sensor element **150**(**2**), the adjacent sensor element **150**(**3**) may also be used. As shown here, while sensor element **150**(**2**) is being operated, the third sensor element **150**(**3**) is connected by the switch module **212** such that the first conductive element **250**(**1**) of the third sensor element **150**(**3**) is connected to the shield signal source, while the second conductive element **250**(**2**) of the third sensor element **150**(**3**) is connected to the ground **248**.
(248) During operation in the first mode **1802** of a sensor element **150** that comprises the two conductive elements **250**(**1**)-(**2**), the adjacent sensor element **150** that also comprises two conductive elements **250**(**1**)-(**2**) may be operated in this way.
(249) In the second mode **1804** while operating the first sensor element **150**(**1**), the switch module **212** connects the conductive elements **150**(**1**)-(**3**) as discussed with regard to FIG. **18**.
(250) While in the second mode **1804** and operating the first sensor element **150**(**1**), in the implementation shown in FIG. **19** the switch module **212** may also connect the capacitive signal source to a first conductive element **250**(**1**) of an adjacent sensor element **150**(**2**), and the shield signal source to the second conductive element **250**(**2**) of the second sensor element **150**(**2**).
(251) While in the second mode **1804** and operating the first sensor element **150**(**1**), the switch module **212** may connect the next adjacent sensor element, such as sensor element **150**(**3**) such that the first conductive element **250**(**1**) of the third sensor element **150**(**3**) is connected to the shield signal source, while the second conductive element **250**(**2**) of the third sensor element **150**(**3**) is connected to the ground **248**.
(252) FIG. **19**B depicts at **1950** conductive element configurations for a second conductive element arrangement in the first mode **1802** and the second mode **1804**, according to some implementations. The configurations associated with these modes may be implemented at least in part by the apparatus control module **132**, or portions thereof as shown.
(253) The configuration depicted comprises conductive elements **250** arranged in a variation of those described with respect to FIGS. **3**A and **3**B. In particular, a compensation sensor element **372** that comprises four conductive elements **250**(**1**)(**1**)-(**1**)(**4**) and sensor elements **150** that comprise three conductive elements **250**(**2**)(**1**)-(**2**)(**3**).
(254) The compensation sensor element **372** has four conductive elements **250**(**1**)(**1**)-(**1**)(**4**) while the sensor elements **150**(**1**)-(**2**) have three conductive elements **250**(M)(**1**)-(M)(**3**). In some implementations the fourth conductive element **250**(**4**) may comprise a second assembly as described above.
(255) In the first mode **1802** while operating the first sensor element **150**(**1**), the switch module **212** connects the conductive elements **250**(**1**)-(**3**) as discussed with regard to FIG. **18**A.
(256) While in the first mode **1802** and operating the first sensor element **150**(**1**), the switch module **212** may connect the conductive elements **250**(**1**)(**1**)-(**1**)(**4**) of an adjacent compensation sensor element **372** to the ground **248**.
(257) While in the first mode **1802** and operating the first sensor element **150**(**1**), the switch module **212** may connect the conductive elements **250**(**3**)(**1**)-(**3**)(**3**) of an adjacent second sensor element **150**(**2**) to the ground **248**.
(258) At **1952** in the second mode **1804** while operating the compensation sensor element **372** in a first variation, the switch module **212** connects the conductive elements **250**(**1**)(**1**)-(**1**)(**4**) as discussed with regard to FIG. **18**B at **1842**.

(259) In the second mode **1804** while operating the compensation sensor element **372** in the first variation, the switch module **212** may connect adjacent conductive elements to the ground **248**.
(260) FIG. **19**C depicts at **1970** conductive element configurations for a second conductive element arrangement in the second variation of the second mode **1804**, according to some implementations. The configurations associated with these modes may be implemented at least in part by the apparatus control module **132**, or portions thereof as shown.
(261) At **1956** in the second mode **1804** while operating the compensation sensor element **372** in a second variation, the switch module **212** connects the conductive elements **250**(**1**)(**1**)-(**1**)(**4**) as discussed with regard to FIG. **18**B at **1854**.
(262) In the second mode **1804** while operating the compensation sensor element **372** in the second variation, the switch module **212** may connect adjacent conductive elements **250** to the ground **248**.
(263) FIG. **20** depicts at **2000** conductive element configurations for a second conductive element **250** arrangement in the first mode **2002** and the second mode **2004**, according to some implementations.
(264) The configuration depicted comprises conductive elements **250** arranged as described with respect to FIG. **5**. The sensor element **150**(**1**) has three conductive elements **250**(**1**)-(**3**) while other sensor elements **150**(**2**)-(**3**) have two conductive elements **250**(**1**)-(**2**). The first conductive elements **250**(**1**) have a width W**51** that is greater than the widths W**53** of the second conductive elements **250**(**2**) and the third conductive elements **250**(**3**). As described above, in some implementations the third conductive elements **250**(**3**) may comprise a second assembly.
(265) In the first mode **1802** while operating the first sensor element **150**(**1**), the switch module **212** connects the conductive elements **150**(**1**)-(**3**) as discussed with regard to FIG. **18**.
(266) While in the first mode **1802** and operating the second sensor element **150**(**2**), the switch module **212** connects a sensor terminal of the capacitive sensor module **206**, such as connected to a capacitive signal source, to the first conductive element **250**(**1**) of the second sensor element **150**(**2**). The switch module **212** also connects the second conductive element **250**(**2**) of the second sensor element **150**(**2**) to the ground **248**.
(267) In the second mode **1804** while operating the first sensor element **150**(**1**), the switch module **212** connects the conductive elements **150**(**1**)-(**3**) as discussed with regard to FIG. **18**.
(268) While in the second mode **1804**, the sensor elements **150**(**2**)-(**3**) having only the first conductive element **250**(**1**) and the second conductive elements **250**(**2**) may not be utilized, or may be connected to ground **248** as shown.
(269) FIG. **21** is a flow diagram **2100** of a process for using the apparatus **102** to determine compensation data, according to some implementations. The process may be implemented at least in part by the apparatus control module **132**.
(270) At **2102**, at a first time, first capacitance data **116** is determined using the first set of sensor elements **150** operated in a first mode **1802**. For example, the apparatus control module **132** determines load capacitance data **254**(**1**).
(271) At **2104**, at a second time, second capacitance data **116** is determined using the first set of sensor elements **150** operated in the second mode **1804**. For example, the apparatus control module **132** determines reference capacitance data **256**(**1**).
(272) At **2106** first compensation data is determined based on the first and second capacitance data **116**. For example, the compensation module **258** may determine the first compensation data based on a difference between the reference capacitance data **256**(**1**) and the load capacitance data **254**(**1**). In another example, the compensation module **258** may determine the first compensation data based on a lookup table, mathematical function, or ratio, that is representative of a relationship between the load capacitance data **254**(**1**) and the reference capacitance data **256**(**1**).
(273) At **2108**, based on the first capacitance data **116** and the first compensation data, first compensated capacitance data is determined. For example, the compensation module **258** may multiply the load capacitance data **254**(**1**) by a correction factor to determine the first compensated capacitance data. The capacitance data **116** may comprise the compensated capacitance data.
(274) In implementations for which the sensor element **150** is not associated with a third conductive element **250**(**3**), the compensation data for an adjacent or proximate sensor element **150** that does include the three conductive elements **250**(**1**)-(**3**) may be used to determine the compensated capacitance data for that sensor element **150**.
(275) The compensation data and the associated compensated capacitance data may be determined before or after the determination of capacitance data **116**. For example, the apparatus control module **132** may determine the compensation data and provide as output the compensated capacitance data for each measurement, as depicted in this implementation.
(276) At **2110**, at a third time, third capacitance data **116** is determined using the first set of sensor elements **150** operated in the first mode **1802**. For example, the apparatus control module **132** determines load capacitance data **254**(**2**).
(277) At **2112**, at a fourth time, fourth capacitance data **116** is determined using the first set of sensor elements **150** operated in the second mode **1804**. For example, the apparatus control module **132** determines reference capacitance data **256**(**2**).
(278) At **2114** second compensation data is determined based on the third and fourth capacitance data **116**. For example, the compensation module **258** may determine the first compensation data based on a difference between the reference capacitance data **256**(**2**) and the load capacitance data **254**(**2**). In another example, the compensation module **258** may determine the first compensation data based on a lookup table, mathematical function, or ratio, that is representative of a relationship between the load capacitance data **254**(**2**) and the reference capacitance data **256**(**2**).
(279) At **2116**, based on the third capacitance data **116** and the second compensation data, second compensated capacitance data is determined. For example, the compensation module **258** may multiply the load capacitance data **254**(**2**) by a correction factor to determine the second compensated capacitance data.
(280) At **2118**, based on the first compensated capacitance data and the second compensated capacitance data, first change data is determined.
(281) At **2120**, based on the first change data, interaction data **128** indicative of an interaction with a lane **104** associated with the first set of sensor elements **150** is generated.
(282) FIG. **22** is a flow diagram **2200** of a process for combining operation of the sensor elements **150** to determine proximity of an object, according to some implementations. The process may be implemented at least in part by the apparatus control module **132**.
(283) The apparatus **102** may be operated in a low-power mode. Use of the low-power mode may be desired to allow extended operational times using a battery, to minimize heat dissipation such as when the apparatus **102** is located within a controlled climate such as a freezer or refrigerated case, to minimize power consumption, and so forth.
(284) While in a low-power mode **2230**, the apparatus **102** may consume less electrical power than while operating in an active power mode **2232**. For example, the low power mode **2230** may utilize a lower sampling rate than the active power mode **2232**, may operate the onboard electronics at a lower clock frequency, may use a different scan pattern of the sensor elements **150** as described with regard to this figure, and so forth.
(285) The apparatus control module **132** may be operated as a proximity sensor, such that the switch module **212** connects a plurality of the conductive elements **250** to provide a relatively large area self-capacitance sensor that is responsive to the approach of a user **108** or other object. Once an object has been detected, the apparatus **102** may transition to the active power mode **2232**.
(286) At **2202** first capacitance data **116** is determined using a plurality of conductive elements **250** that are electrically coupled together. For example, the switch module **212** may connect the first conductive element **250**(**1**) of the first set of sensor elements **150** and the first conductive element **250**(**1**) of the second set of sensor elements **150** to the capacitive signal source of the capacitive sensor module **206**. The capacitive sensor module **206** may then be operated to determine the raw capacitance data **252**. In some implementations, the compensation process described above with regard to FIG. **21** may be used, and compensated capacitance data may be determined.
(287) In another example, the first conductive element **250**(**1**) of a first sensor element **150**(**1**) may be connected to the capacitive signal source of the capacitive sensor module **206** and the first conductive element **250**(**1**) of a second sensor element **150**(**2**) may be connected to the ground **248**.

(288) In some implementations, the apparatus **102** may be operating in the low power mode **2230** at **2202**. The capacitive sensor module **206** may operate in a high sensitivity mode, such that the approach of the user **108** or another object may be detected.
(289) At **2204** second capacitance data **116** is determined using the plurality of conductive elements **250** that are electrically coupled together. In some implementations, the compensation process described above with regard to FIG. **21** may be used, and compensated capacitance data may be determined.
(290) At **2206** a first difference is determined between the first capacitance data **116** and the second capacitance data **116**.
(291) At **2208** the first difference is determined to exceed a threshold value. This may be considered indicative of an approach of an object to the apparatus **102**, such as a user **108**.
(292) In another implementation, operations **2204-2208** may be omitted, and the first capacitance data **116** may be compared to a threshold value. If a value of the first capacitance data **116** exceeds the threshold value, this may be considered indicative of an approach of an object to the apparatus **102**, such as the user **108**.
(293) Responsive to the determination of an object, the process may transition to the active power mode **2232**. The capacitive sensor module **206** may operate in a low sensitivity mode.
(294) For example, the apparatus **102** may be operated in a proximity detection mode that determines capacitance data **116** at a low sample rate, such as once every 500 milliseconds, and for a specified set of sensor elements **150**, such as all. Upon determining the presence of an object, such as the user **108**, the apparatus **102** may transition from a low power mode **2230** to an active power mode **2232** and proceed to **2210** during which capacitance data **116** is determined at a faster sample rate such as every 100 ms and for sets of sensor elements **150** that are associated with particular lanes **104**.
(295) At **2210** third capacitance data is determined using a first subset of the plurality of conductive elements **250**. For example, the conductive elements **250** associated with a first sensor element **150**(**1**) associated with a first lane **104**(**1**) may be operated.
(296) At **2212** fourth capacitance data is determined using a second subset of the plurality of conductive elements **250**. For example, the conductive elements **250** associated with a third sensor element **150**(**3**) associated with a second lane **104**(**2**) may be operated.
(297) FIG. **23** is a flow diagram **2300** of a process for using a first sensor element **150** proximate to a second sensor element **150** to determine a type of interaction, according to some implementations. The process may be implemented at least in part by the apparatus control module **132**. In some implementations the apparatus **102** may be utilized from the front and the back. For example, the apparatus **102** may be restocked by having items **106** placed to the back, while items **106** are picked from the front. In another example, the apparatus **102** may be accessible by users **108** from the front and back, and items **106** may be interacted with from either. In still other examples, the apparatus **102** may be accessible by users **108** from the left or right sides, and items **106** may be interacted with from either.
(298) In such implementations, additional information as to which side an interaction is taking place may be advantageous in determining the interaction data **128**.
(299) In a first implementation shown at **2350**, the apparatus **102**(**1**) comprises a first sensor element **150**(**1**) that is associated with a first end of an inventory location, such as a lane **104**, and a second sensor element **150**(**2**) that is associated with the inventory location.
(300) In a second implementation shown at **2352**, the apparatus **102**(**2**) comprises a first sensor element **150**(**51**) that is associated with a first end of an inventory location and extends across a width of the apparatus **102**(**2**), and a second sensor element **150**(**52**) is associated with the inventory location. The first sensor element **150**(**51**) has a first long axis that extends along the X axis, while the second sensor element **150**(**52**) has a second long axis that extends along the Y axis, placing the first long axis perpendicular to the second long axis.
(301) In a third implementation shown at **2354**, the apparatus **102**(**3**) comprises a first sensor element **150**(**71**) that is associated with a first end of an inventory location and extends across a width of the apparatus **102**(**3**), a second sensor element **150**(**72**) is associated with the inventory location, and a third sensor element **150**(**73**) that is associated with a second end of the inventory location and extends across the width of the apparatus **102**(**3**).
(302) At **2302** a first sensor element **150**(**1**) that is associated with a first end of the inventory location is determined.
(303) At **2304** a second sensor element **150**(**2**) that is associated with the inventory location is determined. For example, the apparatus control module **132** may retrieve lane configuration data **130** that indicates the first sensor element **150**(**1**) and the second sensor element **150**(**2**) are associated with a particular lane **104**.
(304) At **2306** first capacitance data **116** is determined using the first sensor element **150**(**1**).
(305) At **2308**, based on the first capacitance data **116**, proximity data indicative of an object being proximate to the first sensor element **150**(**1**) at the first time is determined. For example, if the value(s) of the first capacitance data **116** exceeds a threshold value, an object may be deemed to be proximate to the first sensor element **150**(**1**). In another example, if a difference in successively measured capacitance data **116** exceeds a threshold value, an object may be deemed to be proximate to the first sensor element **150**(**1**).
(306) At **2310**, at a second time, second capacitance data **116** is determined using the second sensor element **150**(**2**).
(307) At **2312**, based on the proximity data and the second capacitance data **116**, a type of interaction associated with the second sensor element **150**(**2**) is determined. The type of interaction may be indicative of a relative location of a side of the apparatus **102** that is involved in a change, a pick, or place of an item **106** to the inventory location. For example, the first sensor element **150**(**1**) may be associated with a front of the apparatus **102**. If the proximity data indicates the first sensor element **150**(**1**) detects an object at the first time, and then a change in capacitance at the second sensor element **150**(**2**) is determined at the second time, the type of interaction may be deemed to be associated with the front of the apparatus **102**. In comparison, if the proximity data indicates no object detected at the first time, but a change is detected at the second time, the type of interaction may be associated with the back of the apparatus **102**. In some implementations, the location of the interaction, such as front or back may be further associated with a type of interaction. For example, the apparatus **102** may be restocked by placing items **106** at the back, while users **108** pick items **106** from the front. Based on this information, the type of interaction may be determined.
(308) Given the lane configuration data **130** that associated the second sensor element **150**(**2**) with a particular lane **104**, the location of the interaction may be determined. As a result, the type of the interaction and location of the interaction may be determined. Additional information such as the mathematical sign of the change in capacitance as provided by the capacitance data **116** acquired at successive times may be used to determine whether a pick or place occurred, and in some implementations a count of the items **106** picked or placed.
(309) Embodiments may be provided as a software program or computer program product including a non-transitory computer-readable storage medium having stored thereon instructions (in compressed or uncompressed form) that may be used to program a computer (or other electronic device) to perform processes or methods described herein. The computer-readable storage medium may be one or more of an electronic storage medium, a magnetic storage medium, an optical storage medium, a quantum storage medium, and so forth. For example, the computer-readable storage media may include, but is not limited to, hard drives, optical disks, read-only memories (ROMs), random access memories (RAMs), erasable programmable ROMs (EPROMs), electrically erasable programmable ROMs (EEPROMs), flash memory, magnetic or optical cards, solid-state memory devices, or other types of physical media suitable for storing electronic instructions. Further, embodiments may also be provided as a computer program product including a transitory machine-readable signal (in compressed or uncompressed form). Examples of transitory machine-readable signals, whether modulated using a carrier or unmodulated, include, but are not limited to, signals that a computer system or machine hosting or running a computer program can be configured to access, including signals transferred by one or more networks. For example, the transitory machine-readable signal may comprise transmission of software by the Internet.
(310) Separate instances of these programs can be executed on or distributed across any number of separate computer systems. Thus, although certain steps have been described as being performed by certain devices, software programs, processes, or entities, this need not be the case, and a

variety of alternative implementations will be understood by those having ordinary skill in the art.
(311) Additionally, those having ordinary skill in the art will readily recognize that the techniques described above can be utilized in a variety of devices, environments, and situations. Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the claims.

Claims

1. An apparatus comprising: a cover layer; a first sensor element comprising: a first conductive element located beneath the cover layer; a second conductive element located beneath the first conductive element; a first electrical insulator between the first conductive element and the second conductive element; a third conductive element; a second electrical insulator between the second conductive element and the third conductive element; a third electrical insulator beneath the third conductive element; a fourth conductive element, wherein the third electrical insulator is between the third conductive element and the fourth conductive element; and a fourth electrical insulator beneath the fourth conductive element; first electronics to operate the first sensor element to determine first capacitance data; and a first set of conductors that couple the first conductive element, the second conductive element, the third conductive element, and the fourth conductive element to the first electronics.
2. The apparatus of claim 1, wherein one or more conductive elements of the first sensor element have a conductor coverage ratio that is less than one.
3. The apparatus of claim 1, further comprising: a second sensor element comprising: a fifth conductive element located beneath the cover layer; a sixth conductive element located beneath the fifth conductive element; a fifth electrical insulator between the fifth conductive element and the sixth conductive element; a seventh conductive element; a sixth electrical insulator between the sixth conductive element and the seventh conductive element; an eighth conductive element; a seventh electrical insulator between the seventh conductive element and the eighth conductive element; and an eighth electrical insulator beneath the eighth conductive element; a second set of conductors that couple the fifth conductive element, the sixth conductive element, the seventh conductive element, and the eighth conductive element to the first electronics; and the first electronics to operate the second sensor element to determine second capacitance data.
4. The apparatus of claim 1, further comprising: a second sensor element comprising: a fifth conductive element located beneath the cover layer; a sixth conductive element located beneath the fifth conductive element; a fifth electrical insulator between the fifth conductive element and the sixth conductive element; and a sixth electrical insulator beneath the sixth conductive element; the first electronics to operate the second sensor element to determine second capacitance data; a second set of conductors that couple the fifth conductive element and the sixth conductive element to the first electronics; and wherein: the cover layer, the first conductive element, the first electrical insulator, the second conductive element and the second electrical insulator comprise a first assembly; the third conductive element, the third electrical insulator, the fourth conductive element, and the fourth electrical insulator comprise a second assembly; and the second assembly is joined to the first assembly.
5. The apparatus of claim 1, wherein the cover layer is at least 1 mm thick and comprises an electrical insulator.
6. The apparatus of claim 1, wherein the cover layer is at least twice a thickness of the first electrical insulator.
7. The apparatus of claim 1, wherein the first sensor element is shaped as a first triangle; and further comprising a second sensor element shaped as a second triangle, the second sensor element comprising: a fifth conductive element located beneath the cover layer; a sixth conductive element located beneath the fifth conductive element; a fifth electrical insulator between the fifth conductive element and the sixth conductive element; a seventh conductive element; a sixth electrical insulator between the sixth conductive element and the seventh conductive element; and a seventh electrical insulator beneath the seventh conductive element; the first electronics to operate the second sensor element to determine second capacitance data; a second set of conductors that couple the fifth conductive element, the sixth conductive element, and the seventh conductive element to the first electronics; and wherein a first side of the first triangle is proximate to a second side of the second triangle.
8. The apparatus of claim 1, wherein the first sensor element has a first end, a second end, and a first long axis that extends through the first end and the second end; and further comprising: a second sensor element arranged proximate to the first end and having a second long axis that is perpendicular to the first long axis, the second sensor element comprising: a fifth conductive element located beneath the cover layer; a sixth conductive element located beneath the fifth conductive element; a fifth electrical insulator between the fifth conductive element and the sixth conductive element; and a sixth electrical insulator beneath the sixth conductive element; the first electronics to operate the second sensor element to determine second capacitance data; and a second set of conductors that couple the fifth conductive element and the sixth conductive element to the first electronics.
9. An apparatus comprising: a cover layer; a first sensor element comprising: a first conductive element located beneath the cover layer; a second conductive element located beneath the first conductive element; a third conductive element located beneath the cover layer; and wherein the conductive elements are separated by one or more electrical insulators; first electronics to operate the first sensor element to determine first capacitance data; a first set of conductors that couple the first conductive element, the second conductive element, and the third conductive element to the first electronics; a second sensor element comprising: a fourth conductive element located beneath the cover layer; a fifth conductive element located beneath the fourth conductive element; a sixth conductive element located beneath the fifth conductive element; a seventh conductive element located beneath the sixth conductive element; and a first electrical insulator beneath the seventh conductive element; a second set of conductors that couple the fourth conductive element, the fifth conductive element, the sixth conductive element, and the seventh conductive element to the first electronics; and the first electronics to operate the second sensor element to determine second capacitance data.
10. The apparatus of claim 9, the first sensor element further comprising: an eighth conductive element located beneath the cover layer, wherein a second electrical insulator of the one or more electrical insulators is between the third conductive element and the eighth conductive element; and a third electrical insulator beneath the eighth conductive element; and wherein the first set of conductors couple the eighth conductive element to the first electronics.
11. The apparatus of claim 9, wherein one or more conductive elements of the first sensor element have a conductor coverage ratio that is less than one.
12. The apparatus of claim 9, wherein the first conductive element and the third conductive element are coplanar, the first conductive element has a first width, and the third conductive element has a second width that is less than the first width.
13. The apparatus of claim 9, wherein; the cover layer, the first conductive element, the second conductive element, and a first portion of the one or more electrical insulators comprise a first assembly; the third conductive element and a second portion of the one or more electrical insulators comprise a second assembly; and the second assembly is joined to the first assembly.
14. The apparatus of claim 9, wherein the first conductive element has a first width, the second conductive element has a second width, and the third conductive element has a third width, and further wherein the first width is greater than the second width and the third width.
15. The apparatus of claim 9, wherein the cover layer is at least 2 mm thick and comprises an electrical insulator.
16. The apparatus of claim 9, wherein the first sensor element has a first end, a second end, and a first long axis that extends through the first end and the second end; and wherein the second sensor element is arranged proximate to the first end and has a second long axis that is perpendicular to the first long axis.
17. An apparatus comprising: a cover layer; a first sensor element comprising: a first conductive element located beneath the cover layer; a second conductive element located beneath the first conductive element; and a third conductive element located beneath the cover layer; wherein the

conductive elements are separated by one or more electrical insulators; and wherein the first sensor element has a first end, a second end, and a first long axis that extends through the first end and the second end; first electronics to operate the first sensor element to determine first capacitance data; a first set of conductors that couple the first conductive element, the second conductive element, and the third conductive element to the first electronics; a second sensor element arranged proximate to the first end and having a second long axis that is perpendicular to the first long axis, the second sensor element comprising: a fourth conductive element located beneath the cover layer; and a fifth conductive element located beneath the fourth conductive element; the first electronics to operate the second sensor element to determine second capacitance data; and a second set of conductors that couple the fourth conductive element and the fifth conductive element to the first electronics.
18. The apparatus of claim 17, wherein one or more conductive elements of the first sensor element have a conductor coverage ratio that is less than one.
19. The apparatus of claim 17, wherein the cover layer is at least 1 mm thick and comprises an electrical insulator.
20. The apparatus of claim 17, wherein the cover layer is at least twice a thickness of the one or more electrical insulators.